



Review article

Central fatigue theory and endurance exercise: Toward an interoceptive model

Terry McMorris^{a,b,c,*}, Martin Barwood^d, Jo Corbett^a^a Department of Sport and Exercise Science, Faculty of Science, University of Portsmouth, Guildhall Walk, Portsmouth PO1 2ER, United Kingdom^b Department Sport and Exercise Science, Institute for Sport, University of Chichester, College Lane, Chichester, West Sussex PO19 6PE, United Kingdom^c Department of Psychology, Faculty of Health and Life Sciences, Northumbria University, 10 Northumberland Road, Newcastle-upon-Tyne NE1 8ST, United Kingdom^d Department of Sport, Health and Nutrition, Leeds Trinity University, Brownberrie Lane 12, Horsforth LS18 5HD, United Kingdom

ARTICLE INFO

Keywords:

5-hydroxytryptamine
 Insula cortex
 Anterior cingulate cortex
 Dorsolateral prefrontal cortex
 Ventrolateral prefrontal cortex
 Ventromedial prefrontal cortex
 Substantia nigra pars compacta
 Ventral tegmental area
 Locus coeruleus
 Dopamine
 Norepinephrine
 Motivation

ABSTRACT

We propose a model of exercise-induced central fatigue based on interoception and motivation. Predictions of the expected sensory feedback are fed forward by the dorsolateral (DL) prefrontal cortex (PFC) to the anterior insula cortex (AIC). During exercise, the AIC receives feedback from lamina I lateral spinothalamic and nucleus tractus solitarius medullothalamic pathways. The feedback is compared to the predictions in order to generate a current awareness state, which is forwarded to the anterior cingulate cortex (ACC), ventromedial (VM) PFC and lateral (L) PFC. The LPFC integrates the information and makes a decision as to whether to continue or stop. The decision is dependent upon interaction with the substantia nigra pars compacta and ventral tegmental area dopamine (DA), and locus coeruleus (LC)-norepinephrine (NE) systems. Phasic activation of DA and NE neurons appears to be necessary for maintenance of goal-related action but the VMPFC and ACC, which project to the LC, induce tonic NE activity when the rewards are thought to be not worth the cost thus fatigue is perceived.

1. Introduction

Fatigue is an encompassing term which includes impairments in the ability to perform physical tasks or to produce muscle force as well as sensations that relate to tasks being more difficult or taking more effort than expected (Taylor and Gandevia, 2008). Fatigue can originate from both ‘peripheral’ and ‘central’ mechanisms. The nature of peripheral fatigue, i. e. force reductions occurring due to processes distal to the neuromuscular junction (Carroll et al., 2017), has received much attention (see Allen et al., 2008; Kent-Braun et al., 2012; Debold et al., 2016 for reviews) and is relatively well understood. However central fatigue, which has been defined as the inability of the brain to maintain the drive necessary to produce the desired force or power output (Davis and Bailey, 1996) and refers to processes within motoneurons and the central nervous system (Carroll et al., 2017), is perhaps less well-understood. In particular, and as noted previously (Taylor and Gandevia, 2008), the study of exercise-related ‘central’ fatigue has, historically, focused on the performance of the motor system, and less on the more cognitive aspects of fatigue. Yet, anecdotal evidence, often in the form of “ghosted autobiographies” of athletes, shows that the role of cognition or central fatigue during endurance exercise has been known to be

very important for many years. Its relationship with peripheral fatigue is less clear.

In sub-section 1.1, we briefly describe the role of the brain in motor control, while in Section 2 we outline and critique the major neurochemical theories of central fatigue. In Section 3 we examine the major psychophysiological theories. These theories have been developing over a number of years although comparatively recent research into the neuroanatomy of feedback from peripheral afferents to the brainstem and higher centers of the brain (Craig, 1995; 2003; 2004a; 2004b) has led to some interesting developments with regard to the mechanisms thought to account for central fatigue (see Hilty et al., 2011a, 2011b; Robertson and Marino, 2016). In Section 4, we elaborate on these mechanisms and suggest how the nature of Craig’s (2002, 2015) interoception theory provides a viable explanation of the causes of central fatigue. To Craig, interoception is not merely a neuroanatomical phenomenon but is an emotional response which is heavily influenced by motivation. Therefore, we include in our model an account of the role of catecholamines in motivation, particularly with regard to maintaining goal directed behavior, but also how this is affected by exercise-induced alterations in brain catecholamines concentrations. The interaction between exercise and brain catecholamines concentrations does

* Corresponding author at: 63 Four Winds Court, Hartlepool TS26 0LP, United Kingdom.
 E-mail address: t.mcmorris@chi.ac.uk (T. McMorris).

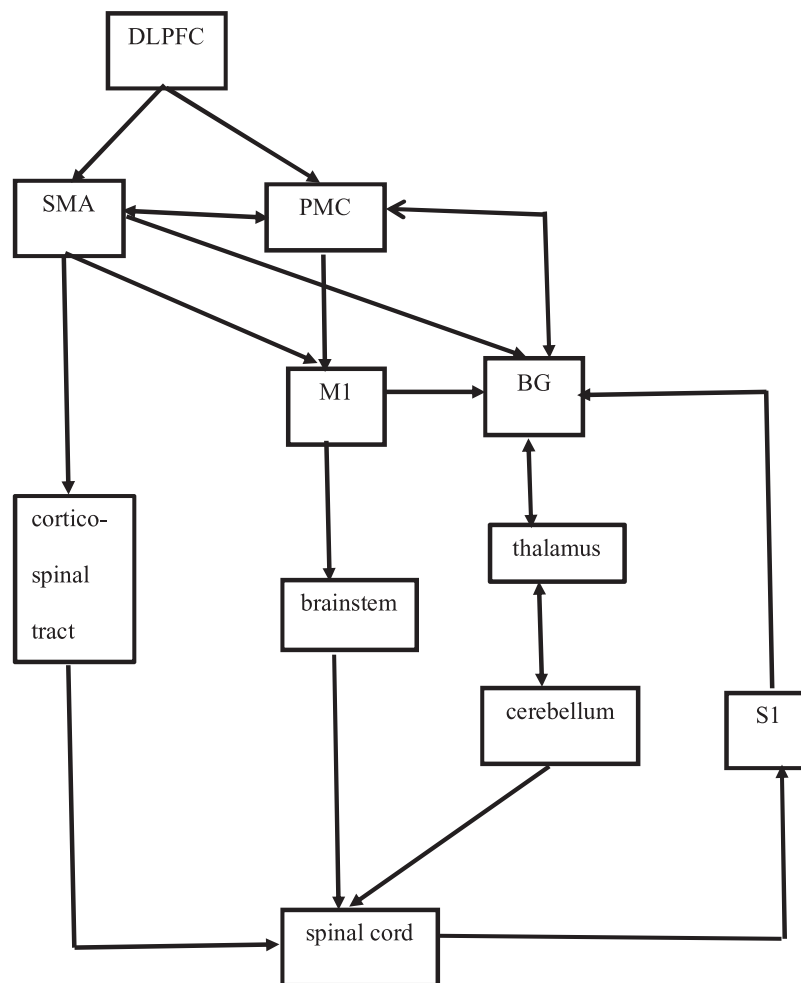


Fig. 1. Schematic of the roles of the different regions of the brain during motor control.

DLPFC dorsolateral prefrontal cortex: SMA supplementary motor area: PMC pre-motor cortex: M1 primary motor cortex: BG basal ganglia: S1 somatosensory cortex.

not only affect motivation but also the whole process of central fatigue, as dopamine (DA) and norepinephrine (NE) are vital for activation of the prefrontal cortex (PFC), which is thought to control central fatigue (Klass et al., 2016; Robertson and Marino, 2016).

1.1. Overview of the role of the brain in motor control

Fig. 1 provides a schematic of the roles of the different regions of the brain during motor control. The decision to undertake physical exercise is thought to be initiated by the dorsolateral (DL)PFC and may come from internal or external input. The supplementary motor area (SMA) is mostly concerned with internal factors, while the premotor cortex (PMC) is more readily activated by external stimuli. The SMA and PMC make direct connections with the spinal cord via the cortico-spinal tract or indirectly via connections with the primary motor cortex (M1) and brainstem. The pre-motor and motor regions of the brain, and indeed most of the cerebral cortex, feedforward to the basal ganglia (BG) and cerebellum, forming the cerebral-BG-thalamo-cerebellar pathway. Moreover, regions projecting forward to the cerebellum, via this pathway, receive ascending feedback from the cerebellum. These closed-loop connections between the BG and cerebellum play a major role in motor control (Doya, 2000; Shadmehr and Krakauer, 2008).

The feedforward to the cerebellum is in the form of corollary discharge. The somatosensory cortex also receives corollary discharge from M1 (Enoka and Stuart, 1992). Corollary discharge informs the cerebellum and somatosensory cortex of the expected sensory consequences of performing the movement correctly. When movements are

performed very slowly, corollary discharge can be compared with the actual sensory feedback and changes to the action can be made. Afferent feedback to the somatosensory cortex is via the dorsal column-lemniscal pathway, which consists of large dynamiter sensory fibers from the skin, muscles and joints, and the ventral spinothalamic pathway, which originates in lamina V and lamina VII. In most cases, feedback arrives too late for the individual to use it to control on-going movement. The problem is overcome due to the plasticity of the cerebellum. Through experience, the cerebellum learns to anticipate errors and alter subsequent movements rather than try to alter the on-going movement. In other words, it develops internal templates that predict the sensory outcomes of motor commands and corrects motor commands through internal feedback. This results in smooth coordinated actions. Although this is probably the most common mechanism used by the cerebellum to control movement, it is generally agreed that when possible, mixed feedforward and feedback control is the best strategy, combining the efficiency of anticipatory action with feedback control (Herreros and Verschure, 2013).

The above briefly outlines the brain mechanisms involved in motor control in non-fatiguing actions. However, as we will see in the following sections, there is some disagreement between theorists as to the part played by corollary discharge and afferent feedback to the somatosensory cortex, with regard to central fatigue. In the next two sections, we examine the major central fatigue theories.

2. Neurochemical theories of central fatigue

2.1. Serotonin hypothesis

It has long been acknowledged that ‘central’ factors play a role in the development of fatigue (see Gandeia, 2001 for seminal review) and that during sustained muscle contraction, the maximal effort that can be achieved voluntarily is less than that which can be achieved when the muscle is activated directly by electrical stimulation of the motor nerve (e.g. Asmussen, 1979). These observations led the Oxford University biochemist, and marathon runner, Eric Newsholme, to propose that the neuromodulator 5-hydroxytryptamine (5-HT), also known as serotonin, might be a prime candidate for explaining central fatigue (Newsholme et al., 1987). Given the mechanisms by which Newsholme et al. (1987) perceived 5-HT to act on the brain during fatiguing exercise, they termed the theory, the tryptophan-5-HT-central fatigue theory. It is now more commonly known as the serotonin theory.

Newsholme et al. (1987) were aware of the commonly held belief that 5-HT induces lethargy and sleepiness. Moreover, they were also aware that the precursor of 5-HT, tryptophan, more readily crosses the blood-brain barrier during exercise than at rest. Tryptophan is found in plasma either bound to albumin or unbound. Unbound or free tryptophan readily crosses the blood-brain barrier. During exercise, free fatty acids displace tryptophan from binding with albumin, therefore there is an increase in free tryptophan. This crosses into the brain and forms 5-HT (Blomstrand, 2006; Hawkins et al., 2006). Some albumin-bound tryptophan also crosses the blood-brain barrier probably due to a dissociation mechanism that takes place at the surface of the brain capillary endothelium (Pardridge, 1998). However, Fernstrom and Fernstrom (2006) questioned the role of free fatty acids and unbinding of tryptophan from albumin as the cause for exercise-induced increases in 5-HT concentrations but did agree that exercise likely induces increased brain concentrations of 5-HT.

Having crossed the blood-brain barrier, tryptophan is hydroxylated to 5-hydroxytryptophan (5-HTP), under the influence of tryptophan hydroxylase. It is further broken down by aromatic amino acid decarboxylase (AADC) into 5-HT. This process takes place in the raphe nuclei. This is the only place that 5-HTP is thought to be found. 5-HT is stored in vesicles, mainly the parafollicular cells of the thyroid (Lefebvre et al., 2001). Tryptophan hydroxylase is the rate-limiting enzyme for 5-HT synthesis and is not fully saturated under normal conditions, therefore increases in brain concentrations of tryptophan will facilitate 5-HT synthesis. Newsholme et al. (1987) received contemporary support from rodent studies, which demonstrated significant increases in brain 5-HT concentrations and/or concentrations of the 5-HT metabolite 5-hydroxyindoleacetic acid (5-HIAA) following exercise (Blomstrand et al., 1989; Chaouloff et al., 1986; 1987). More recent rodent studies have shown similar results (Caperuto et al., 2009; Chen et al., 2008; Chennaoui et al., 2001; Gomez-Merino et al., 2001; Langfort et al., 2006; Meeusen et al., 1996; Meeusen et al., 2001a,b; Meeusen and De Meirleir, 1995). Moreover, Blomstrand et al. (2005) examined the brain uptake of tryptophan during prolonged exercise (3 h at 200 ± 7 W, on a cycle ergometer) in humans by calculating the arterial-venous difference ($a-v$ diff) multiplied by plasma flow. They found large increases in cerebral uptake. However, as reported in a short but succinct recent review (Meeusen and Roelands, 2017), results from studies with humans, in which diet was manipulated to attempt to alter brain concentrations of tryptophan, have failed to support the 5-HT hypothesis.

The methods used in these nutrition studies were tryptophan ingestion and the increase of dietary uptake of branched-chain amino acids (BCAA). Tryptophan ingestion should lead to increased brain tryptophan concentrations, as it crosses the blood-brain barrier. Theoretically, this should induce earlier onset of fatigue. Dietary uptake of BCAA results in a lowering of brain concentrations of tryptophan. BCAA and tryptophan utilize the same blood-brain barrier facilitative

transporter L1 (Hawkins et al., 2006), therefore following increased BCAA intake, there is more competition for the free-tryptophan (i.e. unbound from albumin) to access L1 in order to cross the blood-brain barrier. As a result, there will be less 5-HT synthesized by the raphe nuclei. Hence the inhibitory effect of 5-HT should be attenuated. That the studies failed to support the serotonin hypothesis led Davis and Bailey (1996) to examine a possible interaction between 5-HT and DA as causes of central fatigue.

2.1.1. 5-hydroxytryptamine-dopamine interaction effects

Based on the positive roles of DA in arousal, motor control and motivation, Davis and Bailey (Bailey et al., 1993; Davis and Bailey, 1996; Davis et al., 2000) hypothesized that a low ratio of brain 5-HT to DA favors improved performance but a high ratio of 5-HT to DA would lead to a lowering of arousal, loss of motor coordination and motivation. This they claimed was what caused central fatigue (Davis et al., 2000). This was based on the findings of Bailey et al. (1993), who examined the effect of exercise to exhaustion on the turnover of 5-HT and DA in the midbrain, striatum, hypothalamus and hippocampus of rodents. The authors included the effect of the use of selected drugs, but here we report only results for the placebo trials as our interest is in central fatigue in normal rather than pharmaceutically engineered situations. Measures were taken from rodents sacrificed after one hour as well as those sacrificed following exhaustion. 5-HT and its metabolite 5-HIAA demonstrated significant increases from rest to one hour in all four brain regions measured and further increases in midbrain and striatum following exhaustion. This was similar to the results of Blomstrand et al. (1989) and Chaouloff et al., 1986, using similar methods. Results for DA and its metabolite 3,4 dihydroxyphenylacetic acid (DOPAC) were more complex, however. DA was increased after one hour in the midbrain and striatum only. At exhaustion it had returned to resting levels in the midbrain but remained higher than at rest in the striatum, although this was significantly lower than concentrations after one hour. In the midbrain, DOPAC was significantly higher than at rest after both one hour and exhaustion. Concentrations following one hour and exhaustion did not differ significantly from one another. In the striatum, after one hour, DOPAC was significantly higher than at rest and following exhaustion, while concentrations at exhaustion were significantly greater than at rest. There were no significant differences in the hypothalamus and hippocampus.

In the Abstract to their article, the authors stated “Brain dopamine (DA) and 3,4-dihydroxyphenylacetic acid (DOPAC) were higher at 1 h of exercise ($P < 0.05$) but were similar to resting levels at fatigue” (Bailey et al., 1993, p. 3006) and it is the implications of this comment which have tended to be used as support for a monoamine hypothesis of central fatigue. As we saw in the previous paragraph, the data suggest that the situation was not that straightforward and there is evidence of increased turnover of DA in the midbrain and striatum at exhaustion. Furthermore, Heyes et al. (1988) provided contradictory evidence regarding DA turnover at exhaustion. These authors examined the effect of exercise to exhaustion on concentrations of DA, DOPAC and another metabolite of DA, 4-hydroxy 3-methoxyphenylacetic acid also known as homovanillic acid (HVA), as well as 5-HT and 5-HIAA in the striatum, brainstem, and hypothalamus of rodents. They found increased concentrations of DA, DOPAC and HVA in the striatum and brainstem of exhausted rats. There were no significant effects in the hypothalamus. 5-HIAA concentrations were increased in striatum but 5-HT concentrations were unaffected in all three brain regions. Blomstrand et al. (1989) demonstrated increased DA and 5-HT in the brainstem and hypothalamus, and 5-HIAA in the hippocampus and striatum of mice, at exhaustion. More recently, Hu et al. (2015) have shown increased extracellular DA and 5-HT in the subthalamic nucleus of rats at exhaustion. Observation of the data supplied by Hasegawa et al. (2008) suggests that DA concentrations in the hypothalamus at exhaustion were similar to those at rest, although the authors did not statistically compare concentrations at rest to those at exhaustion. Moreover,

pharmacological and electrophysiological research into the interactions between 5-HT and DA neurons in non-exercise conditions has produced somewhat contradictory results (see Barnes and Sharp, 1999; Fink and Göthert, 2007; Olijslagers et al., 2006, for reviews), which may explain the issues with the 5-HT-DA interaction during exercise (see below).

That there is not strong support for the Bailey et al. (1993) theory may be due to the effects of different 5-HT or serotonergic receptors on DA synthesis and release. 5-HT modulates DA release in the substantia nigra pars compacta (SNc) and ventral tegmental area (VTA), and in their projection areas, the striatum and, the nucleus accumbens (NAc) and PFC respectively (Fink and Göthert, 2007). The 5-HT_{1A}, 5-HT_{1B}, 5-HT_{2A} and 5-HT_{2C} serotonergic receptors, in particular, interact directly and/or indirectly with DA neurons (Olijslagers et al., 2006) in these brain regions. Pharmacological and electrophysiological studies with rodents tend to show that the 5-HT₁ serotonergic receptors have an indirect facilitative effect on DA synthesis and release in these areas (Barnes and Sharp, 1999; Fink and Göthert, 2007). 5-HT₁ receptors couple to G_i/G_o guanosine triphosphate (GTP)-binding proteins and inhibit activation of the second messenger cyclic adenosine monophosphate (cAMP), thus dampening the effects of neuronal activity. However, 5-HT_{1B} receptors are probably located on inhibitory γ -aminobutyric acid (GABA) interneurons rather than on DA projection terminals (Barnes and Sharp, 1999), while 5-HT_{1A} receptors are probably located at the presynaptic terminals of GABAergic neurons (Katsurabayashi et al., 2003). Therefore, they negatively affect the release of the inhibitory neurotransmitter GABA, resulting in an indirect facilitation of DA release. 5-HT_{1A} neurons in the dorsal raphe nucleus open K⁺ channels but do not appear to affect the inhibition of cAMP, therefore they also have a direct facilitative effect on DA release (Clarke et al., 1996).

The situation with the 5-HT₂ family is less clear. These receptors couple to G_q/G₁₂ GTP-binding proteins and activate phospholipase C, resulting in an excitatory effect on 5-HT (Barnes and Sharp, 1999). However, they have been shown to have an inhibitory effect on DA release in the medial (m)PFC. This is because of neurons that are located on inhibitory GABAergic interneurons, which probably activate GABA_B receptors thus inhibiting DA release (Santiago et al., 1995). The same process does not occur in the striatum where 5-HT_{2A} neurons do not appear to affect DA release. Why and how this occurs is unsure and a matter of some debate (see Fink and Göthert, 2007). Despite the inhibitory effect of 5-HT_{2A} receptors on mPFC DA release, 5-HT_{2A} receptors in the mPFC have been shown to facilitate DA release in the VTA (Bortolozzi et al., 2005; Puig et al., 2003). This may be due to activation of 5-HT_{2A} neurons that are located on excitatory, glutamergic pyramidal projection neurons rather than on GABAergic receptors. These neurons project to the VTA area (Bortolozzi et al., 2005). It appears that the effects of 5-HT_{2A} neurons on DA release depend on the brain region involved and the localized positioning of the neurons. The situation with regard to 5-HT_{2C} is far more clear. These receptors are localized on GABAergic interneurons in the dorsal raphe nucleus, therefore inducing an indirect tonic inhibitory control on the DA release in both the NAc and the striatum.

The contradictory empirical results outlined above and the pharmacological and electrophysiological literature concerning the interaction between 5-HT receptors and DA release question the assertion that central fatigue is the result of 5-HT inducing reductions in brain DA release and metabolism. Meeusen and colleagues (e.g., Meeusen et al., 2001a,b; Meeusen and De Meirleir, 1995) decided that the interaction between exercise, 5-HT, DA and central fatigue was incomplete and effects of the neurotransmitter NE, also known as noradrenaline, required investigation.

2.1.2. 5-hydroxytryptamine-catecholamines effects

DA and NE are part of the catecholamines family of neurohormones, along with epinephrine (Epi: also known as adrenaline), and are known to interact with one another to control activity in many regions of the brain (Arnsten, 2009; Gioanni et al., 1998). As a result, Meeusen's

group have undertaken a large amount of research using systematic reviews, research with rodents and more recently pharmacological studies with humans, in order to determine the possible roles of 5-HT, DA and NE in central fatigue. In this sub-section, we examine their work and begin with a summary of their reviews into the effects of acute exercise on brain concentrations of 5-HT, DA and NE in animals.

In a recent article, McMorris (2016) summarized Meeusen and colleagues reviews of animal studies (Meeusen et al., 2001a,b; Meeusen and De Meirleir, 1995) as demonstrating increased DA concentrations, particularly in the brainstem and hypothalamus, during and immediately following acute exercise. Moreover, recent research has supported these results by showing increased DA concentrations in the hypothalamus (Kitaoka et al., 2010) and hippocampus (Goekint et al., 2012). Perhaps more importantly, Meeusen and colleagues demonstrated increased concentrations of DOPAC and HVA, particularly in the brainstem and hypothalamus. However, they found that the effect of acute exercise on whole brain concentrations of NE in animals has shown either a decrease or no significant effect. On the other hand, Kitaoka et al. (2010) demonstrated increased NE concentrations in the hypothalamus, but Goekint et al. (2012) found no significant effect on NE concentrations in the hippocampus. However, Meeusen's reviews showed that animal studies have demonstrated increases in brain concentrations of the NE metabolite 3-methoxy 4-hydroxyphenylglycol (MHPG) in most brain regions. One can conclude that acute exercise does induce increased turnover of DA and NE in the brain, but with some regional variations.

The studies reviewed, however, were in sub-maximal intensity exercise, while our interest is in the effects of exhaustion. We have seen evidence for increases of DA and its metabolites at exhaustion in some brain regions in the previous sub-section, so here we will examine research into concentrations of NE at exhaustion. Only five studies have examined this (Barchas and Freedman, 1963; Blomstrand et al., 1989; Cicardo et al., 1986; Hasegawa et al., 2008; Moore and Lariviere, 1964). Where drugs were used in these studies, we report only the results for the placebo groups. In two of the experiments (Barchas and Freedman, 1963; Cicardo et al., 1986), whole brain concentrations of NE were decreased in the heat (23 °C) but not at a moderate temperature (15 °C). Moore and Lariviere (1964) showed significant decreases of NE in the heat (23 °C and 37 °C), however they did not examine effects in moderate temperatures. Blomstrand et al. (1989) found increased concentrations of NE in the striatum of mice but not in the brainstem, hypothalamus and hippocampus. Although not statistically comparing rest to exhaustion, Hasegawa et al. (2008) showed that NE concentrations in the hypothalamus at exhaustion were similar to those at rest, in a moderate temperature (18 °C). MHPG was not measured in any of the studies, so it is difficult to comment on turnover, although Cicardo et al. believed that their results were due to increased turnover. Given the limited research outlined above, we need to examine the neurochemistry literature on the interaction between 5-HT and NE, before we can discuss the possibilities of exercise-induced increases in 5-HT brain concentrations resulting in depletion of brain NE.

The processes by which 5-HT affects NE release are very similar to those affecting DA release. 5-HT_{1A} receptors have been shown to stimulate NE release particularly in the hippocampus (Done and Sharp, 1992; Hajo's-Korcsok and Sharp, 1996), frontal cortex (Hajo's-Korcsok and Sharp, 1996), hypothalamus (Suzuki et al., 1995), and VTA (Chen and Reith, 1995). 5-HT_{1A} receptors are probably located at the presynaptic terminals of GABAergic neurons (Katsurabayashi et al., 2003). They are coupled to G_i/G_o GTP-binding proteins and inhibit activation of the second messenger cAMP, thus inhibiting GABA release. As GABA is an inhibitor, this has the net effect of increasing NE release. 5-HT_{2C} receptors have an indirect inhibitory effect on NE release particularly in the PFC and locus coeruleus (LC). They are probably localized on inhibitory GABAergic interneurons and so induce an indirect tonic inhibitory control on NE release (Gobert et al., 2000). The effect of 5-HT_{2A} receptors is somewhat questionable. Gobert et al. raised doubts

about any effect at all, but Szabo and Blier (2001) demonstrated an inhibitory effect on NE release from the LC. This, they argued, occurred because although the 5-HT_{2A} receptors are excitatory, they are probably located on GABA terminals, originating from the prepositus hypoglossi nucleus. The excitatory effect of the 5-HT_{2A} neurons leads to an increase of the degree of activation of GABA_A receptors and thus a decrease in NE neuron firing. However, a recent study of central and peripheral neuromuscular function found that GABA_B receptors in the motor cortex were more directly linked with the etiology of incremental peripheral and central fatigue (Goodall et al., 2018). Specifically, an extended cortical silent period was evident with increasing peripheral fatigue, which is associated with GABA_B activation, whereas an extended short-interval intracortical inhibition ratio, a marker of GABA_A activation, did not occur. Long-lasting K⁺-dependent stimulation-induced inhibitory postsynaptic potentials specific to GABA_B activation are thought to be accountable for this result by contrast to the characteristic short-lasting CL⁻-independent response of the GABA_A pathway (McCormick, 1992). Goodall et al. claimed that their results and those of previous researchers strongly suggest that GABA_B receptors require a higher GABA concentrations or longer exposure to GABA than do GABA_A receptors for activation. As we saw in 2.1.1, GABA_B receptors inhibit DA activation (Santiago et al., 1995). Thus both DA and NE can be indirectly inhibited by 5-HT₂ receptors. These data offer a plausible link between peripheral and central neuromuscular fatigue, and cognitive fatigue. Nevertheless, given the research outlined above, the situation with regard to exhaustion is far from clear. In order to attempt to clarify the situation, below we examine the results of pharmacological studies in humans.

2.1.3. Pharmacological studies in humans

Research examining the effects of DA and/or NE reuptake inhibitors or agonists, on time to exhaustion in rodents, shows somewhat equivocal results (Gerald, 1978; Hasegawa et al., 2008; Heyes et al., 1988; Kalinski et al., 2001). However, Roelands and Meeusen (2010) cast some doubt on the strength of evidence with humans in temperate conditions. Administration of 5-HT reuptake inhibitors to humans has generally demonstrated no significant effect (Meeusen et al., 2001a,b; Parise et al., 2001; Piacentini et al., 2002; Strachan et al., 2004), while others have shown time to exhaustion to be shortened (Davis et al., 1993; Wilson and Maughan, 1992). Meeusen et al. (1997) demonstrated no significant effect of a 5-HT_{2C} antagonist on performance. Moreover, DA reuptake inhibitors have tended to show no significant effect (Onus et al., 2016; Piacentini et al., 2004; Roelands et al., 2008b; Watson et al., 2005), although a significant increase in time to exhaustion was demonstrated in one study (Swart et al., 2009). Similarly, NE reuptake inhibitors have shown no significant effect in some studies (Onus et al., 2016; Piacentini et al., 2002; 2004; Watson et al., 2005) but significant negative effects have been demonstrated in others (Klass et al., 2012; 2016; Roelands et al., 2008a). Notably the doses used in studies in which a negative effect of NE reuptake was demonstrated were higher than in the other studies. Jacobs and Bell (2004) administered the α_1 -adrenoceptor agonist modafinil and demonstrated a significant increase in time to exhaustion. Although this appears to contradict the findings reported above, it probably does not. Modafinil is thought also to inhibit GABA release in the cerebral cortex, which indirectly results in increased DA levels (Ferraro et al., 1996). We should note that the results, which we outline above, are all from studies undertaken in temperate conditions. Studies examining performance in the heat (30 °C) have provided different results.

DA and DA/NE reuptake inhibitors induced improved behavioral performance in two studies (Roelands et al., 2008b; Watson et al., 2005) but not in a third (Onus et al., 2016). However, in all three studies, thermoregulation was negatively affected by the DA/NE agonists without a negative effect on exercise performance. The authors argued that motivation would appear to have been significantly, positively affected by increased DA and NE uptake and therefore able to

overcome the desire to stop exercising. This is a logical conclusion given the nature of the study designs. Interestingly, Onus et al. also presented evidence to show that in the cooler condition, the drugs acted peripherally to alter the twitch characteristics of skeletal muscle. The authors claimed that this latter factor could explain the lack of improvements shown for DA/NE agonists in the temperate conditions in their and other studies. Contrary to these results, the NE reuptake inhibitor reboxetine induced a decrement in outcome performance in the heat (Roelands et al., 2008a). It is possible that high concentrations of NE in the ventromedial (VM) PFC and ACC resulted in tonic firing of LC-NE neurons, which reduces drive to continue the present activity and search for alternatives (Aston-Jones and Cohen, 2005; see 2.1.4 and 4.1.1 for discussion of the effects of tonic versus phasic firing of NE neurons).

2.1.4. Issues with monoamine hypotheses

Roelands and Meeusen (2016) stated that the serotonin theory alone can not explain central fatigue but probably plays a large part in it. There is some evidence from pharmacological studies and from the neurochemical literature concerning the interaction between 5-HT neurons and catecholamines release to support the notion that 5-HT, DA and NE do, in fact, interact with one another but the interactions depend on which 5-HT receptors are involved, the regions of the brain in which they are found and more importantly their specific location on GABAergic or glutamergic neurons. Thus, the interactions are complex and are also affected by DA and NE interactions with one another and interactions between catecholamines and the Hypothalamic-Pituitary-Adrenal (HPA) axis hormones. However, we begin our critique by commenting on some much more fundamental issues concerning the interaction between acute exercise and brain catecholamines concentrations.

The basic premise of serotonin theory is that exercise induces increased brain concentration of 5-HT and these inhibit the release of DA and NE in the brain (Meeusen et al., 2006). This is seen as being important because the catecholamines neurotransmitters are vital in the whole process of motor control from the decision to act, and the on-line control of that action through to and including the decision to stop (central fatigue itself). Inhibition of DA and NE during this process is seen as being negative and leading to early fatigue. As we have seen above, the inhibitory effects of 5-HT on catecholamines release are probably not as great as originally thought, which may explain why the serotonin theory is not the definitive answer to central fatigue. However, two fundamental issues with regard to the roles of brain catecholamines concentrations have been ignored in the development of this theory. Firstly, exercise per se induces increased brain concentrations of catecholamines (see McMorris, 2016); and secondly, catecholamines tend to have an inverted-U effect on control of brain activity particularly in the PFC (Arnsten, 2011; Cools and D'Esposito, 2011).

The idea that acute exercise induces increased brain catecholamines concentrations was originally based on the argument that although catecholamines do not readily cross the blood-brain barrier, if circulating concentrations were high, the blood-brain barrier would be compromised (Cooper, 1973). This was supported by the research of Samorajski and Marks (1962), who found that in mice, high concentrations of catecholamines were able to cross the blood-brain barrier in the median eminence at the base of the hypothalamus and in the anterior pituitary gland. More recently, it has been shown that peripherally circulating Epi and NE activate β -adrenoceptor chemoreceptors on the vagus nerve. The excitatory neurotransmitter glutamate mediates synaptic communication between the vagal afferents and the nucleus tractus solitarii (NTS), allowing noradrenergic cells in the NTS, which project to the LC, to stimulate NE synthesis and release to other parts of the brain (Miyashita and Williams, 2006). It was thought that this would occur following the lactate threshold (LT), which marks the beginning of an exponential rise in peripheral blood lactate concentrations (Podolin et al., 1991) and hence significant increases in

circulating Epi and NE. Indeed, Soya and associates (Ohiwa et al., 2006; Soya et al., 2007), experimenting with rodents, have shown that acute exercise above LT induces c-Fos expression, which is indicative of neuronal activity, in what Dahlstroem and Fuxe (1964) termed the A1 and A2 noradrenergic neurons in the NTS. Moreover, Ohiwa et al. also demonstrated that exercise below LT can induce similar changes. Activation of the NTS at sub-LT intensities is extremely unlikely to have been the result of circulating plasma catecholamines activating β -adrenoceptors on the vagus nerve. However, afferent signals from mechanoreceptors, or more accurately stretch receptors, in the heart and lungs, are fed back to the NTS via the vagus nerve (Berthoud and Neuhuber, 2000; Moor et al., 2005; Mravec, 2006). Similarly, arterial baroreceptors provide feedback, concerning blood pressure, to the NTS via the glossopharyngeal and vagus nerves (Kougiyas et al., 2010). Heart rate, tidal volume and blood pressure begin to increase immediately that exercise begins (Watson, 1974), and the feedback allows the hypothalamus to initiate activation of the sympathoadrenal system, culminating in the synthesis and release of catecholamines, in anticipation of increased exercise intensity (Mason et al., 1973; McMorris et al., 2009). Thus, it is not surprising to see c-Fos expression in A1 and A2 neurons in the NTS even prior to the LT.

Rodent studies have also shown that acute exercise induces c-Fos expression in adrenergic C1 neurons in the rostral ventrolateral medulla (Abbott et al., 2013; Barna et al., 2012). C1 neurons project to the LC (Abbott et al., 2012; Guyenet et al., 2013) and are the most likely to establish glutamergic synapses with the LC, although A1 and A2 neurons also innervate the LC (Holloway et al., 2013; Rinaman, 2011). C1, A1 and A2 neurons also have an indirect effect on the LC via projections to the hypothalamus (Guyenet et al., 2013; Rinaman, 2011), which in turn projects to the LC (Aston-Jones et al., 1986), the main source of NE in the brain.

A1, A2, A5 and A6 or LC neurons also project to the VTA (Mejías-Aponte et al., 2009), where they activate α_1 -adrenoceptors. Grenhoff and associates (Grenhoff, and Svensson, 1993; Grenhoff et al., 1993) have shown that stimulation of the α_1 -adrenoceptors by NE release from the LC potentiates the firing of DA neurons in the VTA. This is probably due to α_1 -adrenoceptor activation inducing enhanced glutamate release, which affects the excitability of DA neurons (Velásquez-Martinez et al., 2012). Also, these noradrenergic neurons, along with the adrenergic C1 neurons, project to the retrorubral field (RRF) in the reticular formation and stimulate DA activation there (Rinaman, 2011). The VTA and RRF have projections to the PFC.

Given the literature outlined above, it is not surprising to find rodent studies demonstrating exercise-induced increases in brain catecholamines and their metabolites (Meeusen and De Meirleir, 1995; Meeusen et al., 1997; Meeusen et al., 2001a,b) and this will undoubtedly reduce the effects of 5-HT inhibition of DA and NE release in the brain. Research into serotonin theory has been based on the hypothesis that 5-HT inhibition of DA and NE release would have a negative effect on brain activity, however there is much evidence to show that the catecholamines neurotransmitters affect brain activity in an inverted-U manner (Arnsten, 2011; Clatworthy et al., 2009; Cools and D'Esposito, 2011), which may well mean that 5-HT inhibition of DA and NE concentrations may, in fact, be beneficial to the brain by helping maintain optimal levels of catecholamines. Below, we outline the mechanisms by which catecholamines and their receptors induce an inverted-U effect and return to how this might affect central fatigue in 4.1.1.

The roles of DA and NE, and their receptors have been extensively examined in the PFC and it is this region that provides the largest evidence for an inverted-U effect. The dopaminergic D_1 -receptor is common in the PFC. D_1 -receptors couple to G_s and G_{olf} GTP-binding proteins and stimulate cAMP activation, which amplifies the effects of neuronal activity. When DA concentrations are moderate, D_1 -receptors dampen neural 'noise' by increasing GABAergic interneuron release which inhibits firing to non-preferred stimuli (Gorelova et al., 2002).

Similarly, when NE concentrations are moderate, the high affinity α_{2A} -adrenoceptors are activated. These are coupled to Gi/Go proteins and activation inhibits adenylyl cyclase activity, and closes hyperpolarization-activated cyclic nucleotide-gated (HCN) channels, which increases the strength of neural signaling in the preferred direction, i.e. enhances the strength of the signal (Wang et al., 2007). Hence D_1 -receptors and α_{2A} -adrenoceptors working together strengthen the signal to noise ratio. However, as catecholamines concentrations increase to levels thought to be found in an individual reaching exercise-induced fatigue, the position changes because during exercise high concentrations of NE are thought to activate the lower affinity α_1 - and β -adrenoceptors in the PFC (McMorris, 2016). α_1 -adrenoceptors are coupled to Gq/G₁₁ proteins which activate phospholipase C, causing increased Ca^{+} release and protein kinase A activity in the cell. This inhibits intracellular signaling. β -adrenoceptors are coupled with G_s proteins which increase cAMP activity resulting in the activation of protein kinase A and the dampening of neuronal activity. There is also increased activation of D_1 -receptors which increases cAMP activation. Arnsten (2009; 2011) claimed that high levels of cAMP opens HCN channels throughout the dendrite, thus weakening network inputs from all directions. This would mean that the efficiency of the PFC is greatly diminished.

Moreover, DA and NE neurons both exhibit two different types of firing, tonic and phasic. In DA neurons, interaction between tonic and phasic discharge affects the PFC, and other brain regions, in an inverted-U manner. Phasic activation occurs during moderate tonic firing and is stimulated by salient stimuli. It is inhibited by faster tonic firing. (Bromberg-Martin et al., 2010). The situation is similar with NE neurons. Tonic rates largely fluctuate according to arousal levels and behavioral states. Phasic bursts are triggered by bottom-up input mechanisms involving novel/salient sensory stimuli and top-down decision-making processes (Devliss and Waterhouse, 2011). Phasic firing is most common during moderate tonic discharge and is inhibited by fast tonic discharge (see 4.1.1 for more detail).

The situation is exacerbated by the fact that heavy and long-duration, moderate intensity exercise stimulate the synthesis and release of the HPA hormone cortisol (corticosterone in animals). It is generally thought that exercise needs to be $\geq 80\%$ maximum volume of oxygen uptake (VO_{2MAX}) (De Vries et al., 2000; Hill et al., 2008; McMorris et al., 2009) or to be of at least 45 min s in duration (Bridge et al., 2003; Jacks et al., 2002; Shojaei et al., 2011) to affect plasma and salivary concentrations of cortisol. Cortisol and corticosterone readily cross the blood-brain barrier and rodent studies have shown evidence of acute exercise-induced increases in corticotropin releasing factor (CRF) mRNA expression in the paraventricular neurons (PVN) of the hypothalamus (Hand et al., 2002; Jiang et al., 2004; Kawashima et al., 2004; Timofeeva et al., 2003; Yanagita et al., 2007). Synthesis and release of CRF is the first stage in the synthesis of cortisol/corticosterone. CRF receptors are found on LC neurons (Van Bockstaele et al., 1996) and this results in increased NE release in the PFC (Höglund et al., 2000). Also, glucocorticoids block the transporters on glia that normally remove catecholamines from the extracellular space (Pruessner et al., 2004). Thus, increased HPA activity can increase NE synthesis and release.

Other regions of the brain also show an inverted-U effect of catecholamines activation. Research into the effect of DA on activation of neurons in the striatum demonstrated that pharmacological or iontophoretic administration of DA had a positive effect on human and non-human animals who had low levels of DA at baseline but had a negative effect on those with high levels at baseline (Clatworthy et al., 2009; Vytacil et al., 2014). Unlike with the PFC, where D_1 -receptors are mostly involved, in the striatum the main effect is on D_2 -receptors, which are coupled to Gi/Go GTP-binding proteins and inhibit adenylyl cyclase activity. The nature of the neural discharge, tonic or phasic also affects striatum efficiency particularly when concerning motivation (Seamans and Yang, 2004).

An inverted-U effect was also shown in the ventroposterior medial

thalamus and somatosensory cortex (Devilbiss and Waterhouse, 2004; Devilbiss et al., 2006), however the inverted-U effect is not always demonstrated in these regions. It is thought that in these regions, facilitation of excitatory responses is mainly mediated by α_1 -adrenoceptors, while suppression of evoked-activity is likely a result of NE acting on postsynaptic α_2 - or β -adrenoceptors (Devilbiss and Waterhouse, 2011). However, CRF can induce tonic firing of LC-NE neurons, which results in suppression of signal transmission within the somatosensory thalamus and cortex (Devilbiss et al., 2012) and this appears to reduce detectability of low-intensity stimuli without affecting high-intensity stimuli (Devilbiss and Waterhouse, 2002; Moore, 2004). Arnsten (2011) claimed that this allowed for quick and accurate detection of stimuli which indicate danger.

Although we believe that the failure to take into account the inverted-U effect of catecholamines in the brain, the role of differing monoamine receptors and the tonic versus phasic firing of DA and NE neurons is a weakness in the neurochemical theories, we must acknowledge that evidence for heavy or long-duration, moderate intensity exercise having the same effect on humans as the stress imposed on rodents, during heavy exhausting exercise, is weak. The fact that several authors (Onus et al., 2016; Piacentini et al., 2004; Roelands et al., 2008b; Watson et al., 2005) showed that in temperate conditions, DA reuptake inhibitors did not affect performance and that NE reuptake inhibitors only had an effect when the dose was very high (Roelands et al., 2008a), strongly suggests that maximal intensity exercise in humans does not have the same effects on brain catecholamines as that shown in rodents. This is probably because the rodents need to be forced to exercise at these intensities. Even in sub-maximal exercise with rodents, this has been shown to result in higher concentrations of brain CRF and corticosterone in forced exercise compared to voluntary exercise (Droste et al., 2008). Furthermore, research with animals has shown that in extreme stress, the PFC is taken off-line (Arnsten, 2011), probably due to D_1 -receptors, α_1 - and β -adrenoceptors dampening all neural activity in the PFC. If this occurred in humans, central fatigue would not be under PFC control. Moreover, research examining the effect of acute exercise on cognition, while showing optimal effects during moderate intensity exercise, supplies only limited support for a deterioration in central executive activity, which is dependent on PFC activation, and certainly does not show a complete shut-down as stress studies with rodents suggest (McMorris and Hale, 2012). This is important because the proponents of psychophysiological theories of central fatigue claim that the PFC is key to perception of effort and fatigue, and the decision to abort activity (Noakes et al., 2004; Marcora, 2009). This could only occur if catecholamines concentrations at exhaustion were at or just over the top of the inverted-U.

To summarize this critique, we can say that there is some support for an interaction between 5-HT, catecholamines and HPA hormones during exercise, and that this may be involved in central fatigue. However, alone it most definitely does not provide strong evidence of being the major influence. therefore in the next section, we examine the literature on the major psychophysiological or psychobiological theories of central fatigue.

3. Psychophysiological theories

3.1. 'Central governor' theory

In this sub-section, we examine a theory based in psychophysiology. Noakes and colleagues (Noakes, 2012; Noakes et al., 2004; St. Clair Gibson et al., 2006) have developed what has become known as the 'central governor' or 'central integrative' theory. Founded on the writings of the nineteenth century Italian physiologist, Angelo Mosso (see Di Giulio et al., 2006; Noakes, 2012) as well as the Nobel Prize winning physiologist A. V. Hill (Noakes and Marino, 2009), they claim that the organism attempts to maintain homeostasis, even during exercise, in order to self-protect. During exercise, there is great danger of

homeostasis being violated. The physiological needs during exercise, if not checked, could damage the organism, therefore the brain terminates, or regulates, the activity before homeostasis is threatened (Noakes and Marino, 2009; Noakes, 2012). Indeed, there are empirical data consistent with this assertion. For instance, Kay et al. (2001) have shown that when participants were instructed to perform a 1 minute maximal sprint at 10 minute intervals during 60 minutes of self-paced cycling in a hot-humid environment, power output and the integrated electromyography signal were reduced in sprints 2–5, compared to sprint 1. However, during the final sprint, participants were able to return their power output and the associated integrated electromyography signal to near baseline (sprint 1) values. These data were interpreted as indicating that efferent drive was subconsciously controlled in a manner that maintained a 'muscle reserve' during sprints 2–5, which the participants were able to utilize during the final sprint as part of a regulatory process which served to prevent premature fatigue or physiological damage. Likewise, Amann et al. (2006) demonstrated that 5 km cycling time trial performance was impaired with hypoxia and improved with hyperoxia, relative to normoxia, yet peripheral (quadriceps) fatigue was not different between the conditions. This was interpreted as indicating that the locomotor muscle output is determined to a significant extent by the regulation of central motor output to the working muscle, in order that peripheral muscle fatigue does not exceed a critical threshold.

In examining how this might occur, Noakes and colleagues (Noakes, 2012; Noakes and Marino, 2009; Noakes et al., 2004; St. Clair Gibson et al., 2006) have drawn heavily on Ulmer's (1996) teleo-anticipation theory of psychophysiological feedback. According to Ulmer (1996), efferent signals feedforward to the spinal cord and muscles to initiate action. In turn, afferent signals from the periphery feedback to a central 'black box' programmer or central governor, which modifies the efferent commands. The programmer, however, does not simply compare the initial efferent feedforward with the afferent feedback, but takes into account the expected end-point of the exercise, past experience of similar activity, fitness level, nutritional status, arousal level and many other aspects even motivation levels (Lambert et al., 2005). Thus the individual can teleo-anticipate the necessary changes to be made to the efferent commands. Noakes and colleagues make many references to the importance of knowledge of the end-point of the exercise or the ability to predict the likely end-point (Lambert et al., 2005; Noakes, 2012; Noakes et al., 2004). This appears to be very important for teleo-anticipation but it does not, however, explain why the individual terminates exercise at any given point, this is thought to depend on the person's perception of fatigue.

St. Clair Gibson et al. (2003) saw fatigue as being a conscious sensation (perception is probably terminologically more accurate than the word "sensation") rather than a physiological occurrence. As we saw in the previous paragraph, during exercise afferent feedback is continuously available to the central governor, allowing for changes to be made to the efferent command. St. Clair Gibson and Noakes et al., (2004) argued that although this process takes place at a sub-conscious level, "the subconscious brain informs the conscious brain of an increasing neural effort" (p. 801) as the individual experiences difficulty in maintaining homeostasis at the given intensity. The brain is thought to interpret this as an "increased sensation of fatigue" (p. 801). The central governor uses this information and other feedback including motivation level, to produce a "homeostatically (sic) acceptable exercise intensity" (p. 797). The calculation is guided by knowledge of the end-point of the exercise. If the end-point is not known, the person must draw on past experience and afferent feedback to teleo-anticipate the required output. Interestingly, St. Clair Gibson et al. (2003) draw on the theory of Damasio (1993) from which Craig (2002) developed his interoception model. More pertinently, Rauch et al. (2005) drew on the neuroanatomy aspects of Craig's (2002) interoception theory to propose a mechanism involving feedback from type III and IV afferents to the insula cortex via the lamina I spinothalamicocortical pathway.

Furthermore, they claimed that projections from the insula to the PFC may be used to determine whether to continue at the current pace or to make alterations. Others (Hilty et al., 2011a; 2011b; Robertson and Marino, 2016) also supported the notion of feedback from lamina I afferents via the spinothalamocortical pathway to the insula as being a potential source of information concerning perception of effort and fatigue (see section 4 for more detail).

3.2. Marcora's psychobiological theory

Disagreement with the claims concerning the role of afferent feedback, posited by proponents of the central governor/integrative theory, led Marcora, 2009 to propose an alternative theory which they called psychobiological theory. However, as we will see in this sub-section, there are many similarities between the theories, a point accepted by Pageaux et al. (2014). According to psychobiological theory, there are five key factors involved in central fatigue: "1) Perception of effort; 2) Potential motivation; 3) Knowledge of the distance to cover; 4) Knowledge of the distance covered/remaining; 5) Previous experience/memory of perceived exertion during exercise of varying intensity and duration" (Marcora, 2010, pp. 454–455). While perception of effort is seen as vital by both groups, Marcora's notion of this factor is very different to that of Noakes (2012). Marcora (2009) saw perception of effort as depending not on feedback but rather on corollary discharge, the expected sensory consequences of performing the movement correctly, which is fed forward to the sensory regions of the brain. As the individual tires or for some reason the task becomes more difficult, e.g. running uphill, efferent commands need to be altered in order to meet the demands of the task. The increased effort is consciously detected by the individual but any decisions, as to what to do, are dependent on the other four factors.

Marcora (2010) used the term "potential motivation", rather than simply motivation, as it refers to the amount of effort that the individual is willing to exert in order to be successful in the task. Not surprisingly, it depends on a whole range of factors that determine motive strength at any given point in time. Potential motivation will interact with knowledge of the distance to cover, knowledge of the distance covered/remaining and previous experience/memory of perceived exertion during exercise of varying intensities and durations, in order to decide whether to attempt to continue with the activity or to terminate it. However, it is thought that at this stage, there is conflict between competing responses, i.e. to stop or continue exercising (Pageaux et al., 2014). Several authors (Marcora, 2009; Pageaux et al., 2014; 2015) have argued that this requires activation of the pre-SMA and ACC, and these brain areas have been shown to be activated during exercise.

3.3. Summary of the similarities and differences between the psychophysiological theories

Both theories are in agreement that corollary discharge plays a major role in controlling activity and that it is responsible for setting the initial parameters with regard to expected sensory consequences. To the proponents of the central governor theory (Noakes, 2012; Noakes et al., 2004), afferent feedback provides the brain with the necessary information to be aware when the task is getting difficult. To Marcora (2009), perception of effort is the "conscious awareness of the central motor commands to the locomotor and respiratory muscles" (p. 2061). Marcora does not deny that afferent feedback plays roles in a variety of physiological and perceptual responses, but denies that it has a role in perception of effort. However, there is a great deal of agreement concerning the factors taken into account to determine whether to continue or to terminate activity. Both groups agree that motivation is a key issue and that factors such as distance to be covered and past experience of similar activities are also important in decision making. There is also agreement that when the decision stage is reached, there is conflict

between the "stop" and the "continue" responses. There is also some agreement on which brain regions are involved in this decision.

Proponents of the central governor theory have cited the insula cortex (Hilty et al., 2011a; Noakes, 2012; Williamson et al., 1999), the ventromedial (VM)PFC including the medial orbitofrontal cortex, the lateral (L)PFC, ACC, PMC, SMA and cerebellum (Hilty et al., 2011a; Mehta et al., 2009; Noakes, 2012; Robertson and Marino, 2016; St. Clair Gibson et al., 2003) as interacting with one another to provide information to the central governor to decide whether to continue or stop. Whether it is this interaction that constitutes the central governor or one of these regions acts as the central governor is yet to be decided. Similarly, supporters of Marcora's psychobiological theory cite the ACC (Marcora et al., 2009, 2009; Pageaux et al., 2014; 2015) and insula cortex (Marcora, 2009) but also the pre-SMA (Marcora, 2009; Pageaux et al., 2014; 2015) as key to determine whether to continue or abort the exercise. The main concerns with brain regions in this theory appear to be the roles of these regions in response conflict.

4. Toward an interoceptive model of central fatigue

In sections 2 and 3, we have outlined and critiqued the major neurochemical and psychobiological theories of central fatigue. While we acknowledge that these theories have played a major part in our understanding of this phenomenon, we believe that there are weaknesses that need to be addressed. The neurochemical theories have failed to take into account the differing properties of neuroreceptors and the roles of tonic versus phasic release of DA from the SNc and VTA, and NE from the LC. The inverted-U effect of catecholamines concentrations in the brain have also been ignored but, in fact, may not be as much of a problem as one might think (this is discussed in 4.1.1). The psychophysiological theories highlight the role of corollary discharge from M1 but disagree regarding the roles of feedback to the somatosensory cortex. Nevertheless, they agree on the importance of perception of effort and also motivation. Given these issues, we propose a more holistic model based on Craig's (2000) interoception theory, which is dependent on feedback from the whole body concerning homeostasis, muscular activity, emotion and motivation. Moreover, we emphasize the interaction between exercise-induced increases and motivational increases in DA and NE, and the brain regions involved in decision making with regard to central fatigue.

4.1. Interoception and central fatigue

Sherrington (1948) saw interoception as being the perception of the physiological condition of the viscera but more recently it has been re-defined as the perception of "the physiological condition of the entire body" (Craig, 2002, p. 655). These feelings have not only a sensory, but also an affective, motivational aspect (Bubic et al., 2010; Craig, 2002). The motivation to undertake exercise is thought to be controlled largely by the DLPFC (Spielberg et al., 2012) but with input from other frontal regions, particularly the VMPFC (Szatkowska et al., 2008) and ACC (Kouneiher et al., 2009). We should, however, remember that the default mechanism in humans is to maintain homeostasis, which may result in aborting an action sooner rather than later. The DLPFC also controls initial top down strategies for achieving the goal (Spielberg et al., 2012) and feeds forward a prediction of the expected sensory feedback to the insula cortex (Craig, 2002). During exercise, we believe that the predicted interoceptive feedback will depend on past experience of similar physical activity, the individual's perception of their current fitness level, the subjective interpretations of the importance of the activity and whether or not the person believes that their actions will be assessed or evaluated by significant others. Directly competing against others is also most likely to affect the interoceptive predictions. One would expect that the predictions will differ in competitive situations. Also, in training when the individual is attempting to improve performance levels, predictions will have to change if the individual is

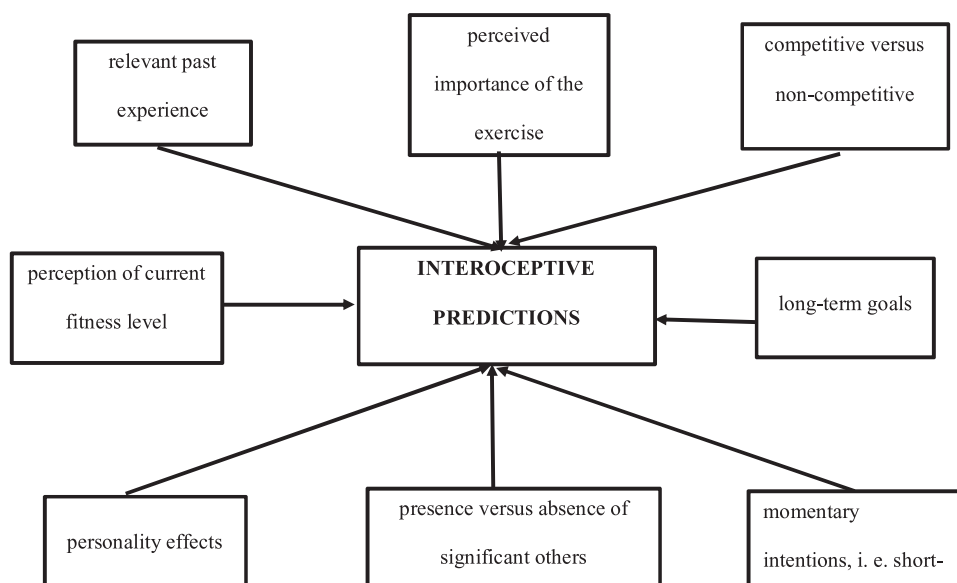


Fig. 2. Factors affecting interoceptive predictions.

to move forward. Furthermore, the predictions will be influenced by the individual's long-term goals, personality, and physical and social development (see Fig. 2).

Based on interoception theory, the individual will make prediction errors even when they have a large amount of experience of the task. As with corollary discharge in the cortico-BG-thalamo-cerebellum pathway, the situation can not be predicted perfectly, therefore there has to be continuous processing of afferent feedback (Craig, 2002). However, the interoceptive feedback pathway (see Fig. 3) is very different to the somatosensory pathway although it does receive input from the somatosensory system. The situation when the person has no or very limited previous experience of the same or similar tasks, as in increasing distance or time during training, raises an issue for making predictions. The prediction errors would be great in initial trials at a new distance or speed but Paterson and Marino (2004) showed that the individual can comparatively quickly establish an “exertion template”, which is altered “on-line” in subsequent trials and over time refined by the person.

The interoceptive feedback pathway begins in the spinal cord. During exercise, the lamina 1 lateral spinothalamocortical pathway is activated. Small-diameter Aδ- and C-type primary afferent fibers, which sense the physiological condition of all tissues of the body and terminate in lamina I of the spinal and trigeminal dorsal horns, relay afferent information in the lateral spinothalamic tract to the main homeostatic integration sites in the brainstem. The latter include regions that also receive vagal and glossopharyngeal afferent feedback via the NTS. The feedback includes information concerning a wide variety of physiological conditions, e. g. temperature, mechanical stress, blood pressure, acidic pH, hypoxia, hypercapnia, hypoglycemia and osmolarity (Craig, 2002, 2015). Both the lamina I lateral spinothalamic and NTS medullothalamic axons terminate in the posterior and basal parts of the ventral medial nucleus of the thalamus (VMpo and VMb respectively) often described as the VMpo + VMb. The VMpo + VMb projects to the insula cortex but before discussing the very important implications of this projection, we should point out two important factors that we have not yet mentioned. Many of the brainstem cites that receive lamina I and NTS inputs provide descending control of the autonomic nervous system and are heavily involved in control of homeostasis. Indeed, Craig (2015) claims that lamina I and NTS connections indicate that their primary function is to provide sensory input to the autonomic, preautonomic and homeostatic cell groups of the spinal cord and brainstem in order to maintain the health of the body. As part of this

process, lamina I and NTS axons terminate on regions of the brainstem which contain concentrations of adrenergic and noradrenergic cells, particularly A1-A2 and A5-A7. As we saw in sub-section 2.1.4, activation of these cells can lead to increased NE release from the LC (Holloway et al., 2013; Rinaman, 2011) but also indirectly increased DA release from the SNc and VTA (Grenhoff, and Svensson, 1993; Grenhoff et al., 1993). We also know that exercise increases activity of these cell groups (Ohiwa et al., 2006; Soya et al., 2007).

Returning to the VMpo + VMb projection to the insula cortex, the afferent information is mapped firstly in the contralateral anterior insula cortex (AIC) and then, by way of a callosal pathway, a lateralized, second-order re-representation is made on the right AIC. This becomes consciously accessible, allowing the individual to make a subjective, affective perception of their physiological and emotional state. The AIC also receives afferent input from the somatosensory cortex (Gu et al., 2013). The AIC compares the interoceptive feedback with the top-down predictions of interoceptive state, which it received from the DLPFC, in order to generate a current awareness state (Gu et al., 2013). This is forwarded to the ACC, VMPFC and LPFC (Craig, 2002).

It is generally thought that the AIC and ACC work in conjunction with one another (Craig, 2002; Medford and Critchley, 2010). Craig described the insula as being a limbic-sensory cortex, because of its association with visceral sensation, and the ACC as a limbic-motor cortex, because of its association with autonomic and emotional control (Devinsky et al., 1995; Mesulam and Mufson, 1982). Similarly, Medford and Critchley (2010) stated that the AIC is responsible for the input and the ACC for the output components of interoception. This interoceptive state is generated by the integrative functions of the AIC and then re-represented in the ACC as a basis for the selection of and preparation for responses to inner or outer events (Craig, 2002; 2015). The ACC is also part of the motivation/reward pathway and contains neurons which are activated by positive motivation (Chudasama et al., 2013). We should note, however, that ACC activation is also linked to aversive processing (Vogt, 2005; Johansen and Fields, 2004).

The insula cortex and ACC have bidirectional projections with the VMPFC, which also has connections with amygdala, striatum, and thalamus (Craig, 2002; Holroyd and Coles, 2002; Singer et al., 2009; Williams and Goldman-Rakic, 1998). It is mainly involved in decision making and motivation. It is seen as playing an important part in the maintenance of goal-directed behavior. The VMPFC, particularly the region known as the orbitofrontal cortex which consists of Brodmann's areas 10, 11 and 47 (Kringelbach, 2005), is thought to evaluate choice

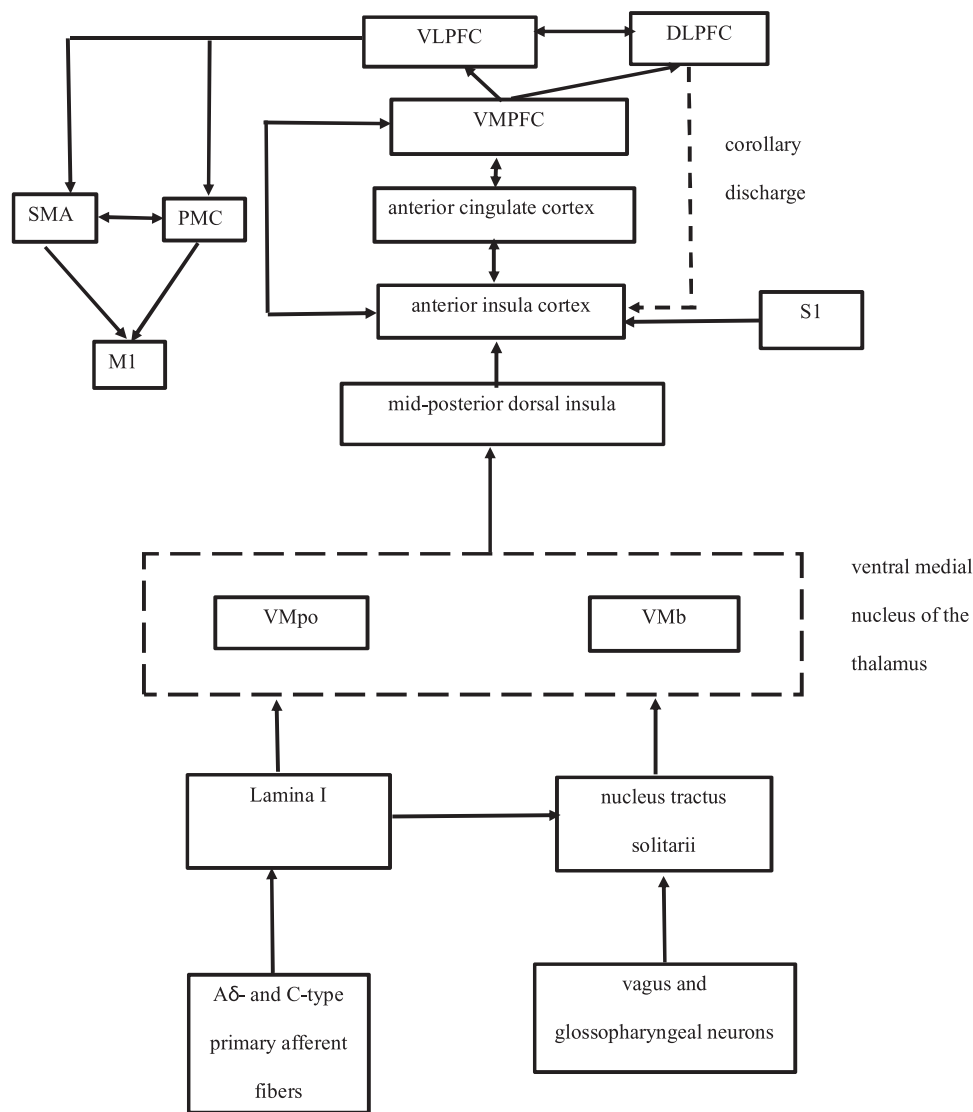


Fig. 3. Schematic of interoceptive feedback pathway.

DLPFC dorsolateral prefrontal cortex: VLPFC ventrolateral prefrontal cortex: SMA supplementary motor area: PMC pre-motor cortex: M1 primary motor cortex: VMPFC ventromedial prefrontal cortex: S1 somatosensory cortex: VMpo posterior ventral medial nucleus of the thalamus: VMb basal ventral medial nucleus of the thalamus.

options and encode outcome expectations (Schoenbaum et al., 2009). The VMPFC projects to the LPFC, as do the insula cortex and ACC (Singer et al., 2009). The LPFC integrates the information received from these regions, as well as information received from the somatosensory cortex via the motor-somatosensory control system (see Fig. 1), and is generally thought to be responsible for making decisions concerning what action to take (Cole et al., 2013; Nee and D'Esposito, 2016). These decisions are based on past experience and the motivational state of the individual. Thus the LPFC has responsibility for continuing the action or stopping it. The LPFC is well connected to the pre-SMA, SMA and PMC, which in turn inform M1 of the chosen response. There is some disagreement as to the exact roles of the DLPFC and ventrolateral (VL)PFC. Fehr and colleagues (Figner et al., 2010; Knoch and Fehr, 2007) claimed that the DLPFC implements inhibition, while Aron et al. (2014) argued that the DLPFC determines task rules or parameters, which we believe could set the level of interoceptive feedback that will indicate the need to stop. In line with Aron et al., we believe that it is the right VLPFC via projections to the pre-SMA, SMA and PMC that initiates the motor action of stopping. Writing specifically about central fatigue, Robertson and Marino (2016) took the sensible option and stated that it

is the LPFC which controls the decision. Nee and D'Esposito (2016) provide evidence to show that when actions are aborted, the hierarchical control in the PFC is led by Brodmann's areas 45 and 46. Area 45 is in the VLPFC, while 46 is in the DLPFC, hence the use of the encompassing term LPFC is very appropriate. However, the efficiency of this interoceptive system is dependent on its interaction with the dopaminergic midbrain systems, the SNc and VTA, and the LC-NE system, particularly with regard to motivation.

4.1.1. Catecholamines

Before and during exercise, feedforward from the DLPFC to the hypothalamus initiates activation of the dopaminergic midbrain systems (Ballard et al., 2011; Bjorklund and Dunnett, 2007; Bromberg-Martin et al., 2010), although other brain regions receiving feedforward from the DLPFC, such as the mPFC, also project to the VTA (Adell and Artigas, 2004). The connections are bidirectional; and the ventromedial SNc and VTA project to the VMPFC, DLPFC and ACC (Holroyd and Coles, 2002; Williams and Goldman-Rakic, 1998). The DA neurons in these PFC regions carry out a number of very important functions with regard to motivation and interoception. However motivation is also

affected by DA neurons in the BG, which also receive input from the SNc and VTA, particularly the dorsal striatum (Haber et al., 2000) including the NAc (Assadi et al., 2009). DA neurons activated in the NAc core are crucial for enabling motivation to overcome response costs such as physical effort; and for an enhancement of general motivation (Cardinal, 2006; Ghods-Sharifi and Floresco, 2010).

However, the actions of DA in these regions and its interaction with NE and 5-HT neurons mean that the effects are not straightforward. We have already seen in 2.1.2 that D₁-like and D₂-like dopaminergic neurons are affected differently by activation due to different effects on the second messenger and/or the location of the receptor with regard to GABAergic and glutamergic neurons. However, Bromberg-Martin et al. (2010) have shown that the nature of firing of the neurons also influences their effects. In a comprehensive review, they presented evidence for DA neurons exhibiting two different types of firing, tonic and phasic. Tonic discharge is controlled by a pacemaker conductance, which is a spontaneous, slow depolarizing membrane current that maintains the basal activity state of the neurons (Grace and Bunney, 1983). Phasic discharge, bursts of rapid firing of neurons, occurs in response to salient stimuli but the amplitude of the response depends on the level of tonic activity (Belujon and Grace, 2015). The more neurons that are firing during tonic discharge, the greater the phasic bursts, although not always as increases in extracellular DA during tonic discharge can attenuate the phasic burst firing-driven transient release (Floresco et al., 2003). This is likely to occur during high levels of stress (Armsten, 2011) and is related to the inverted-U effect. At low levels of stress, firing is slow and tonic, which maintains the baseline level in downstream neural structures in order to ensure normal functioning of the neural circuits. During moderate levels of stress, tonic firing is at a moderate level and phasic bursts rapidly increase or decrease firing of DA neurons for 100–500 ms. This causes large changes in DA concentrations downstream and these last for seconds (Schultz, 1998; 2007). These phasic activations are triggered by reward prediction error. Waelti et al. (2001) showed that during phasic bursts, DA neurons coded the prediction error, which induced greater behavioral and neuronal learning than when predictions were as expected. This may well account for training effects where one would expect greater prediction error as the person moves into uncharted territory. During high levels of arousal or stress, tonic firing is fast with little or no phasic bursts, which has a negative effect on neural activity. Bromberg-Martin et al. (2010) also present evidence to show that one type of DA neurons encode motivational value. These are excited by prediction of reward but inhibited by aversive events. A second type of DA neurons encode motivational salience and are excited by both rewarding and aversive events. According to Bromberg-Martin et al. (2010), motivational value coding neurons project to the VMPFC, NAc shell and the dorsal striatum. Salience coding neurons are claimed to project to the DLPFC and NAc core.

The effects of motivation are also dependent on NE synthesis and release primarily from the LC, which receives direct or indirect input from the hypothalamus, ACC, VMPFC, amygdala and NTS. With regard to motivation, it is most likely that top-down projections from the ACC and VMPFC are important in initiating LC release of NE (Aston-Jones and Cohen, 2005). DLPFC will also have an indirect effect via the hypothalamus. As with DA activity, the different families of adrenoceptors affect activity differently due to effects on the second messenger and also the type of neural firing, tonic or phasic. Tonic discharge is across a range of relatively slow rates (0.1–5.0 Hz), which is stochastically determined. Tonic rates largely fluctuate according to arousal levels and behavioral states. Basically it follows a linear pattern. Phasic discharge consists of brief 10–20 Hz bursts of two to three action potentials that is often, but not always, followed by a sustained suppression of spontaneous activity (200–500 ms) (Akaike, 1982). These bursts are triggered by bottom-up input mechanisms involving novel/salient sensory stimuli and top-down decision-making processes (Devliss and Waterhouse, 2011). Phasic activity appears to be necessary for maintenance of goal-related action and occurs during moderate tonic activity (Aston-Jones

et al., 1999). Thus as with DA neurons, NE neurons are affected in an inverted-U fashion with regard to goal maintenance (but see below). Moreover, research has shown that whereas dopaminergic neurons appear to encode the expected reward and anticipate the effort cost, noradrenergic neurons mobilize resources in order to energize the behavior necessary for successful completion of the task (Bouret et al., 2012; Varazzani et al., 2015). Aston-Jones and Cohen (2005) claim that the VMPFC and ACC, which project to the LC, initiate phasic bursts of NE activation when goal directed behavior is determined to be effective but induce tonic activity when the rewards are thought to be not worth the cost. So as we saw above, phasic activity ensures an inverted-U effect on goal maintenance but tonic activation of NE is believed to be facilitative of searching for alternative goals (Aston-Jones and Cohen, 2005; Usher et al., 1999) when rewards are thought not to be cost worthy. With regard to exercise, we would expect that when this point is reached, the VMPFC and ACC would feedback to the LPFC, which would abort the exercise.

With regard to the inverted-U effect, the changes in tonic and phasic firing that Aston-Jones and Cohen (2005) discuss are taking place in non-exercise situations. As we saw in 2.1.4, during exercise, feedback from the glossopharyngeal- and vagal-NTS pathways results in increased release of DA and NE in the PFC (Miyashita and Williams, 2006), thus there would be increased extracellular DA, which has a negative effect on DA phasic firing, while high levels of NE would induce tonic firing of NE neurons in the PFC. Such action would lead to a breakdown in the efficiency of the PFC which would mean that it would not be able to control central fatigue. The research of Roelands and colleagues (Onus et al., 2016; Piacentini et al., 2004; Roelands et al., 2008b; Watson et al., 2005), who showed that DA reuptake inhibitors did not negatively affect central aspects of fatigue and NE reuptake inhibitors did so only when doses were very high (Klass et al., 2012; 2016; Onus et al., 2016; Piacentini et al., 2002; 2004; Roelands et al., 2008a; Watson et al., 2005), strongly suggests that maximal intensity exercise is not the same as high levels of stress. It would appear that at exhaustion, the concentrations of DA and NE are probably beginning to rise, which would increase tonic release at the expense of phasic firing of neurons. This would lead to the LPFC searching for alternatives to continuing the exercise, probably leading to central fatigue.

4.1.2. Summary

To summarize: our hypothesis differs from previous theories in that it proposes a holistic model which is based on corollary discharge to the insula cortex and feedback from the whole body not just the exercising musculature. It also integrates the roles of brain catecholamines in decision making concerning if and when to stop exercising. Furthermore, it involves the roles of DA and NE in motivation but adds the importance of tonic versus phasic release of DA and NE neurons, and how this impinges on the decisions to continue or stop exercising.

5. Implications for chronic fatigue

Several authors have suggested that chronic fatigue, as in chronic fatigue syndrome (CFS) or multiple sclerosis (MS) or Parkinson's disease (PD) or similar diseases, is a form of central fatigue (e.g. Dobryakova et al., 2015; Fukuda et al., 1994). Dobryakova et al. (2015), in what they termed the dopamine imbalance hypothesis, claimed that chronic fatigue was experienced because communication between the striatum and PFC was inhibited due to disruptions in DA signaling between these regions. Unlike with acute central fatigue, they point to low level concentrations of DA as being responsible for chronic fatigue. However, they also highlight the inverted-U effect and suggest that high pharmacologically-induced concentrations of DA may have a negative effect. Although they state that at present, there is no conclusive evidence for high levels of DA to induce feelings of fatigue in individuals with CFS, MS, PD or any other similar disease.

The research of Salamone and colleagues (e. g. Salamone et al.,

2003; 2012) examining the effect of DA on motivation to pursue effort-related choice behavior suggests a possible role for low levels of DA having a negative effect and possibly inducing feelings of fatigue, particularly chronic fatigue. They showed that DA in the NAc is inhibited by high concentrations of adenosine. In striatopallidal neurons, adenosine A2A receptors are co-expressed with dopamine D₂ receptors. When adenosine concentrations are increased, D₂ activation is decreased. Moreover, striatal adenosine A1 receptors have been shown to have an antagonistic effect on dopamine D₁ receptors (Shen and Chen, 2009). This occurs due to receptor-receptor crosstalk and interactions at the intercellular second messenger systems (Canals et al., 2003; Morelli and Pinna, 2001). As a result they claimed that motivation to carry out effort-related activity would decrease. Chronic fatigue is thought to induce high levels of striatal adenosine (Dantzer et al., 2014). Indeed, caffeine-induced increases in the brain adenosine:DA ratio have been shown to negatively affect physical-endurance performance (Davis et al., 2003) and Pageaux et al. (2014) have argued that it could be a cause of central fatigue.

The interoception model itself could provide an explanation for chronic fatigue. As we have seen above, the insula cortex and ACC receive a wide range of physiological feedback via the lamina 1 lateral spinothalamocortical and NTS medullothalamic pathways. The insula cortex and the ACC are also connected to the amygdala, which store memories of aversive stimuli. This information is forwarded to the VMPFC which also has connections to the amygdala. As we saw above, the VMPFC is thought to evaluate choice options and encode outcome expectations (Schoenbaum et al., 2009). This suggests that aversive memories associated with CFS and other diseases, which are held in the amygdala, will lead to the decision to avoid activity. This will particularly be the case when DA concentrations are low as in CFS, MS, PD and similar diseases (Dobryakova et al., 2015).

6. Conclusion

During exercise it would appear that what we have termed the motor-somatosensory control system, consisting of the cerebral-BG-thalamo-cerebellum pathway controls the actual movements but we believe that the key issue to central fatigue is the interoception control system. The DLPFC feeds predictions of the expected interoceptive feedback, in the form of corollary discharge, to the AIC before the exercise begins. These predictions are based on a variety of inputs, most important of which is probably past experience, however the predictions are heavily affected by emotion and motivation. In training situations or in competition, when the individual expects to have to exercise a higher intensity than they have previously experienced, the prediction is less easy to make than when they intend to reproduce a previous speed or intensity. However, even in these latter situations, the person has to continuously monitor the interoceptive feedback and make decisions concerning whether to continue, slow down, go faster or harder, or to stop. These decisions are heavily determined by the level of motivation and this motivation is under the control of the SNC and VTA-DA, and LC-NE systems.

Although very high levels of stress lead to over-activation of D₁-like receptors, α_1 - and β -adrenoceptors in the PFC, dampening all neural activity in the PFC and thus, in fact, taking the PFC off-line (Arnsten, 2011), this does not appear to happen in humans when exercise is the stressor. However, when the individual reaches exhaustion it may be that the catecholamines concentrations in the PFC are such that discharge moves from phasic firing to tonic. This has been shown to lower the desire to maintain the goal-directed activity and induce the search for alternatives (Aston-Jones and Cohen, 2005), such as to stop exercising. On the other hand, it may simply be that during heavy exercise, the interoceptive feedback is at such a level that the person decides that they can not continue any further, or rather that continuing is not worth the cost required. If the individual thinks that further activity may injure them, they will stop. In other words, the levels of brain

catecholamines concentrations at maximal intensity exercise may be such that the individual can still make an informed choice. They are probably not over the top of the inverted-U or certainly not far over the top, so the PFC is still working optimally or just-sub-optimally.

Funding sources

This research did not receive any specific grant from funding agencies in the public, commercial, or not-for-profit sectors.

References

- Abbott, S.B., Kanbar, R., Bochorishvili, G., Coates, M.B., Stornetta, R.L., Guyenet, P.G., 2012. C1 neurons excite locus coeruleus and A5 noradrenergic neurons along with sympathetic outflow in rats. *J. Physiol.* 590, 2897–2915.
- Abbott, S.B., Depuy, S.D., Nguyen, T., Coates, M.B., Stornetta, R.L., Guyenet, P.G., 2013. Selective optogenetic activation of rostral ventrolateral medullary catecholaminergic neurons produces cardiorespiratory stimulation in conscious mice. *J. Neurosci.* 33, 3164–3177.
- Adell, A., Artigas, F., 2004. The somatodendritic release of dopamine in the ventral tegmental area and its regulation by afferent transmitter systems. *Neurosci. Biobehav. Rev.* 28, 415–431.
- Akaike, T., 1982. Periodic bursting activities of locus coeruleus neurons in the rat. *Brain Res.* 239, 629–633.
- Allen, D.G., Lamb, G.D., Westerblad, H., 2008. Skeletal muscle fatigue: cellular mechanisms. *Physiol. Rev.* 88, 287–332.
- Amann, M., Eldridge, M.W., Lovering, A.T., Stickland, M.K., Pegelow, D.F., Dempsey, J.A., 2006. Arterial oxygenation influences central motor output and exercise performance via effects on peripheral locomotor muscle fatigue in humans. *J. Physiol.* 575, 937–952.
- Arnsten, A.F.T., 2009. Stress signalling pathways that impair prefrontal cortex structure and function. *Nat. Rev. Neurosci.* 10, 410–422.
- Arnsten, A.F.T., 2011. Catecholamine influences on dorsolateral prefrontal cortical networks. *Biol. Psychiatry* 69, e89–e99. <http://dx.doi.org/10.1016/j.biopsych.2011.01.027>.
- Aron, A.R., Robbins, T.W., Poldrack, R.A., 2014. Inhibition and the right inferior frontal cortex: one decade on. *Cognit. Sci.* 18, 177–185.
- Asmussen, E., 1979. Muscle fatigue. *Med. Sci. Sports* 11, 313–321.
- Assadi, S.M., Yu, M., Pantelis, C., 2009. Dopamine modulates neural networks involved in effort-based decision-making. *Neurosci. Biobehav. Rev.* 33, 383–393.
- Aston-Jones, G., Cohen, J.D., 2005. An integrative theory of locus coeruleus-nor-epinephrine function: adaptive gain and optimal performance. *Annu. Rev. Neurosci.* 28, 403–450.
- Aston-Jones, G., Rajkowski, J., Cohen, J., 1999. Role of locus coeruleus in attention and behavioral flexibility. *Biol. Psychiatry* 46, 1309–1320.
- Aston-Jones, G., Ennis, M., Pieribone, V.A., Nickel, W.T., Shipley, M.T., 1986. The brain nucleus locus coeruleus: restricted afferent control of a broad efferent network. *Science* 234, 734–737.
- Bailey, S.P., Davis, J.M., Ahlborn, E.N., 1993. Neuroendocrine and substrate responses to altered brain 5-HT activity during prolonged exercise to fatigue. *J. Appl. Physiol.* 74, 3006–3012.
- Ballard, L.C., Murty, V.P., Carter, R.M., MacInnes, J.J., Huettel, S.A., Adcock, R.A., 2011. Dorsolateral prefrontal cortex drives mesolimbic dopaminergic regions to initiate motivated behavior. *J. Neurosci.* 31, 10340–10346.
- Barchas, J.D., Freedman, D.X., 1963. Brain amines: response to physiological stress. *Biochem. Pharmacol.* 12, 1232–1235.
- Barna, B.F., Takakura, A.C., Moreira, T.S., 2012. Pontomedullary and hypothalamic distribution of Fos-like immunoreactive neurons after acute exercise in rats. *Neuroscience* 212, 120–130.
- Barnes, N.M., Sharp, T., 1999. A review of central 5-HT receptors and their function. *Neuropharmacology* 38, 1083–1152.
- Belujon, P., Grace, A.A., 2015. Regulation of dopamine system responsivity and its adaptive and pathological response to stress. *Proc. Biol. Sci.* <http://dx.doi.org/10.1098/rspb.2014.2516>.
- Berthoud, H.R., Neuhuber, W.L., 2000. Functional and chemical anatomy of the afferent vagal system. *Auton. Neurosci.* 85, 1–17.
- Bjorklund, A., Dunnett, S.B., 2007. Fifty years of dopamine research. *Trends Neurosci.* 30, 185–187.
- Blomstrand, E., 2006. A role for branched-chain amino acids in reducing central fatigue. *J. Nutr.* 136, 544S–547S.
- Blomstrand, E., Møller, K., Secher, N.H., Nybo, L., 2005. Effect of carbohydrate ingestion on brain exchange of amino acids during sustained exercise in human subjects. *Acta Physiol. Scand.* 185, 203–209.
- Blomstrand, E., Perrett, D., Parry-Billings, M., Newsholme, E.A., 1989. Effect of sustained exercise on plasma amino acid concentrations and on 5-hydroxytryptamine metabolism in six different brain regions in the rat. *Acta Physiol. Scand.* 136, 473–481.
- Bortolozzi, A., Diaz-Mataix, L., Scorza, M.P., Celada, C., Artigas, F., 2005. The activation of 5-HT receptors in prefrontal cortex enhances dopaminergic activity. *J. Neurochem.* 95, 1597–1607.
- Bouret, S., Ravel, S., Richmond, B.J., 2012. Complementary neural correlates of motivation in dopaminergic and noradrenergic neurons of monkeys. *Front. Behav. Neurosci.* <http://dx.doi.org/10.3389/fnbeh.2012.00040>.

- Bridge, M.W., Weller, A.S., Rayson, M., Jones, D.A., 2003. Ambient temperature and the pituitary hormone responses to exercise in humans. *Exp. Physiol.* 88, 627–635.
- Bromberg-Martin, E.S., Matsumoto, O., Hikosaka, K., 2010. Coding of task reward value in the dorsal raphe nucleus. *J. Neurosci.* 30, 6262–6272.
- Bubic, A., von Cramon, Y., Schubotz, R.I., 2010. Prediction, cognition and the brain. *Front. Hum. Neurosci.* <http://dx.doi.org/10.3389/fnhum.2010.00025>.
- Canals, M., Marcellino, D., Fanelli, F., Ciruela, F., de Benedetti, P., Goldberg, S.R., Neve, K., Fuxe, K., Agnati, L.F., Woods, A.S., Ferré, S., Lluis, C., Bouvier, M., Franco, R., 2003. Adenosine A2A-dopamine D2 receptor-receptor heteromerization: qualitative and quantitative assessment by fluorescence and bioluminescence energy transfer. *J. Biol. Chem.* 278, 46741–46749.
- Caperuto, E.C., dos Santos, R.V., Mello, M.T., Costa Rosa, L.F., 2009. Effect of endurance training on hypothalamic serotonin concentration and performance. *Clin. Exp. Pharmacol.* 36, 189–191.
- Cardinal, R.N., 2006. Neural systems implicated in delayed and probabilistic reinforcement. *Neural Netw.* 19, 1277–1301.
- Carroll, T.J., Taylor, J.L., Gandevia, S.C., 2017. Recovery of central and peripheral neuromuscular fatigue after exercise. *J. Appl. Physiol.* 122, 1068–1076.
- Chaouloff, F., Laude, D., Merino, D., Serrurier, B., Guezennec, Y., Elghozi, J.L., 1987. Amphetamine and α -methyl-p-tyrosine affect the exercise-induced imbalance between the availability of tryptophan and synthesis of serotonin in the brain of the rat. *Neuropharmacology* 26, 1099–1106.
- Chaouloff, F., Laude, D., Guezennec, Y., Elghozi, J.L., 1986. Motor activity increases tryptophan, 5-hydroxyindoleacetic acid and homovanillic acid in ventricular cerebrospinal fluid of the conscious rat. *J. Neurochem.* 46, 1313–1316.
- Chen, H.L., Lin, L.C., Yu, L., Liu, Y.F., Kuo, Y.M., Huang, A.M., Chuang, J.L., Wu, F.S., Liao, P.C., Jen, C.J., 2008. Treadmill exercise enhances avoidance learning in rats: the role of down-regulated serotonin system in the limbic system. *Neurobiol. Learn. Mem.* 89, 489–496.
- Chen, N.H., Reith, M.E., 1995. Monoamine interactions measured by microdialysis in the ventral tegmental area of rats treated systemically with (9)-8-hydroxy-2-(di-n-propylamino)tetralin. *J. Neurochem.* 64, 1585–1597.
- Chennaoui, M., Drogou, C., Gomez-Merino, D., Grimaldi, B., Fillion, G., Guezennec, C.Y., 2001. Endurance training effects on 5-HT(1B) receptors mRNA expression in cerebellum, striatum, frontal cortex and hippocampus of rats. *Neurosci. Lett.* 307, 33–36.
- Chudasama, Y., Daniels, T.E., Gorrin, D.P., Rhodes, S.E.V., Rudebeck, P.H., Murray, E.A., 2013. The role of the anterior cingulate cortex in choices based on reward value and reward contingency. *Cereb. Cortex* 23, 2884–2898.
- Cicardo, V.H., Carbone, S.E., Carneiro De Rondina, D., Mastronardi, I.O., 1986. Stress by forced swimming in the rat: effects of mianserin and moclobemide on gabaergic-monoaminergic systems in the brain. *Comp. Biochem. Physiol. C* 83, 133–135.
- Clarke, W.P., Yocca, F.D., Maayani, S., 1996. Lack of 5-hydroxytryptamine1A-mediated inhibition of adenylyl cyclase in dorsal raphe of male and female rats. *J. Pharmacol. Exp. Ther.* 277, 1259–1266.
- Clatworthy, P.L., Lewis, S.J.G., Brichard, L., Hong, Y.T., Izquierdo, D., Clark, L., Cools, R., Aigbirho, F.I., Baron, J.-C., Fryer, T.D., Robbins, T.W., 2009. Dopamine release in dissociable striatal subregions predicts the different effects of oral methylphenidate on reversal learning and spatial working memory. *J. Neurosci.* 29, 4690–4696.
- Cole, M.W., Reynolds, J.R., Power, J.D., Repovs, G., Anticevic, A., Braver, T.S., 2013. Multi-task connectivity reveals flexible hubs for adaptive task control. *Nat. Neurosci.* 16, 1348–1355.
- Cools, R., D'Esposito, M., 2011. Inverted-U shaped dopamine actions on human working memory and cognitive control. *Biol. Psychiatry* 69 (12), e113–e125. <http://dx.doi.org/10.1016/j.biopsych.2011.03.028>.
- Cooper, C.J., 1973. Anatomical and physiological mechanisms of arousal with specific reference to the effects of exercise. *Ergonomics* 16, 601–609.
- Craig, A.D., 1995. Distribution of brainstem projections from spinal lamina I neurons in the cat and the monkey. *J. Comp. Neurol.* 361, 225–248.
- Craig, A.D., 2000. Thermosensory activation of insular cortex. *Nat. Neurosci.* 3, 184–190.
- Craig, A.D., 2002. How do you feel? Interoception: the sense of the physiological condition of the body. *Nat. Rev. Neurosci.* 3, 655–666.
- Craig, A.D., 2003. Distribution of trigeminothalamic and spinothalamic lamina I terminations in the cat. *Somatosens. Mot. Res.* 20, 209–222.
- Craig, A.D., 2004a. Distribution of trigeminothalamic and spinothalamic lamina I terminations in the macaque monkey. *J. Comp. Neurol.* 477, 119–148.
- Craig, A.D., 2004b. Lamina I, but not lamina V, spinothalamic neurons exhibit responses that correspond with burning pain. *J. Neurophysiol.* 92, 2604–2609.
- Craig, A.D., 2015. *How Do You Feel? An Interceptive Moment With Your Neurobiological Self*. Princeton University Press, Princeton, NJ.
- Dahlstroem, A., Fuxe, K., 1964. Evidence for the existence of monoamine-containing neurons in the central nervous system. *Acta Physiol. Scand.* 62, 1–55.
- Damasio, A.R., 1993. *Descartes' Error: Emotion, Reason, and the Human Brain*. Putnam, New York.
- Dantzer, R., Heijnen, C.J., Kavelaars, A., Laye, S., Capuron, L., 2014. The neuroimmune basis of fatigue. *Trends Neurosci.* 37, 39–46.
- Davis, J.M., Bailey, S.P., 1996. Possible mechanisms of central nervous system fatigue during exercise. *Med. Sci. Sports Exerc.* 29, 45–57.
- Davis, J.M., Alderson, N.L., Welsh, R.S., 2000. Serotonin and central nervous system fatigue: nutritional considerations. *Am. J. Clin. Nutr.* 72, S73S–S78S.
- Davis, J.M., Bailey, S.P., Jackson, D.A., Strasner, A.B., Morehouse, S.L., 1993. Effects of a serotonin (5-HT) agonist during prolonged exercise to fatigue in humans. *Med. Sci. Sports Exerc.* 25, S78.
- Davis, J.M., Zhao, Z., Stock, H.S., Mehl, K.A., Buggy, J., Hand, G.A., 2003. Central nervous system effects of caffeine and adenosine on fatigue. *Am. J. Physiol. Regul. Integr. Comp. Physiol.* 284, R399–R404.
- Debold, E.P., Fitts, R.H., Sundberg, C., Nosek, T.M., 2016. Muscle fatigue from the perspective of a single crossbridge. *Med. Sci. Sports. Exerc.* 48, 2270–2280.
- De Vries, W.R., Bernardis, N.T.M., De Rooij, M.H., Koppeschaar, H.P.F., 2000. Dynamic exercise discloses different time-related responses in stress hormones. *Psychosom. Med.* 62, 866–872.
- Devliss, D.M., Page, M.E., Waterhouse, B.D., 2006. Locus ceruleus regulates sensory encoding by neurons and networks in waking animals. *J. Neurosci.* 26, 9860–9872.
- Devliss, D.M., Waterhouse, B.D., 2002. Determination and quantification of pharmacological, physiological, or behavioral manipulations on ensembles of simultaneously recorded neurons in functionally related neural circuits. *J. Neurosci. Methods.* 121, 181–198.
- Devliss, D.M., Waterhouse, B.D., 2004. The effects of tonic locus ceruleus output on sensory-evoked responses of ventral posterior medial thalamic and barrel field cortical neurons in the awake rat. *J. Neurosci.* 24, 10773–10785.
- Devliss, D.M., Waterhouse, B.D., 2011. Phasic and tonic patterns of locus coeruleus output differentially modulate sensory network function in the awake rat. *J. Neurophysiol.* 105, 69–87.
- Devliss, D.M., Waterhouse, B.D., Berridge, C.W., Valentino, R., 2012. Corticotropin-releasing factor acting at the locus coeruleus disrupts thalamic and cortical sensory-evoked responses. *Neuropsychopharmacology* 37, 2020–2030.
- Devinsky, O., Morrell, M.J., Vogt, B.A., 1995. Contributions of anterior cingulate cortex to behavior. *Brain* 118, 279–306.
- Di Giulio, C., Daniele, F., Tipton, C.M., 2006. Angelo Mosso and muscular fatigue: 116 years after the first Congress of Physiologists: IUPS commemoration. *Adv. Physiol. Educ.* 30, 51–57.
- Dobryakova, E., Genova, H.M., DeLuca, J., Glenn, R., Wylie, G.R., 2015. The dopamine imbalance hypothesis of fatigue in multiple sclerosis and other neurological disorders. *Front. Neurol.* <http://dx.doi.org/10.3389/fneur.2015.00052>.
- Done, C.J., Sharp, T., 1992. Evidence that 5-HT₂ receptor activation decreases norepinephrine release in rat hippocampus in vivo. *Br. J. Pharmacol.* 107, 240–245.
- Doya, K., 2000. Complementary roles of basal ganglia and cerebellum in learning and motor control. *Cur. Opin. Neurobiol.* 10, 732–739.
- Droste, S.K., de Groote, L., Atkinson, H.C., Lightman, S.L., Reul, J.M., Linthorst, A.C., 2008. Corticosterone levels in the brain show a distinct ultradian rhythm but a delayed response to forced swim stress. *Endocrinology* 149, 3244–3253.
- Enoka, R.M., Stuart, D.G., 1992. Neurobiology of muscle fatigue. *J. Appl. Physiol.* 7, 1631–1648.
- Fernstrom, J.D., Fernstrom, M.H., 2006. Exercise, serum free tryptophan, and central fatigue. *J. Nutr.* 136, S53S–S59S.
- Ferraro, L., Tanganelli, S., O'Connor, W.T., Antonelli, T., Rambert, F., Fuxe, K., 1996. The vigilance promoting drug modafinil increases dopamine release in the rat nucleus accumbens via the involvement of a local GABAergic mechanism. *Eur. J. Pharmacol.* 306, 33–39.
- Figner, B., Knoch, D., Johnson, E.J., Krosch, A.R., Lisanby, S.H., Fehr, E., Weber, E.U., 2010. Lateral prefrontal cortex and self-control in intertemporal choice. *Nat. Neurosci.* 13, 538–539.
- Fink, K.B., Göthert, M., 2007. 5-HT receptor regulation of neurotransmitter release. *Pharmacol. Rev.* 59, 360–417.
- Floresco, S.B., West, A.R., Ash, B., Moore, H., Grace, A.A., 2003. Afferent modulation of dopamine neuron firing differentially regulates tonic and phasic dopamine transmission. *Nat. Neurosci.* 6, 968–973.
- Fukuda, K., Straus, S.E., Hickie, I., Sharpe, M.C., Dobbins, J.G., Komaroff, A., 1994. The chronic fatigue syndrome: a comprehensive approach to its definition and study. International chronic fatigue syndrome study group. *Ann. Intern. Med.* 121, 953–959.
- Gandevia, S.C., 2001. Spinal and supraspinal factors in human muscle fatigue. *Physiol. Rev.* 81, 1725–1789.
- Gerald, M.V., 1978. Effects of (+)-amphetamine on the treadmill endurance performance of rats. *Neuropharmacology* 17, 703–704.
- Ghods-Sharifi, S., Floresco, S.B., 2010. Differential effects on effort discounting induced by inactivations of the nucleus accumbens core or shell. *Behav. Neurosci.* 124, 179–191.
- Gioanni, Y., Thierry, A.M., Glowinski, J., Tassin, J.P., 1998. Alpha1-adrenergic, D1, and D2 receptors interactions in the prefrontal cortex: implications for the modality of action of different types of neuroleptics. *Synapse* 30, 362–370.
- Gobert, A., Rivet, J.M., Lejeune, F., Newman-Tancredi, A., Adhumeau-Auclair, A., Nicolas, J.P., Cistarelli, L., Melon, C., Millan, M.J., 2000. Serotonin(2C) receptors tonically suppress the activity of mesocortical dopaminergic and adrenergic, but not serotonergic, pathways: a combined dialysis and electrophysiological analysis in the rat. *Synapse* 36, 205–221.
- Goekint, M., Bos, I., Heyman, E., Meeusen, R., Michotte, Y., Sarre, S., 2012. Acute running stimulates hippocampal dopaminergic neurotransmission in rats, but has no impact on brain-derived neurotrophic factor. *J. Appl. Physiol.* 112, 535–541.
- Gomez-Merino, D., Béquet, F., Berthelot, M., Chennaoui, M., Guezennec, C.Y., 2001. Site-dependent effects of an acute intensive exercise on extracellular 5-HT and 5-HIAA levels in rat brain. *Neurosci. Lett.* 301, 143–146.
- Goodall, S., Howatson, G., Thomas, K., 2018. Modulation of specific inhibitory networks in fatigued locomotor muscles of healthy males. *Exp. Brain Res.* 236, 463–473.
- Gorelova, N., Seamans, J.K., Yang, C.R., 2002. Mechanisms of dopamine activation of fast-spiking interneurons that exert inhibition in rat prefrontal cortex. *J. Neurophysiol.* 88, 3150–3166.
- Grace, A.A., Bunney, B.S., 1983. Intracellular and extracellular electrophysiology of nigral dopaminergic neurons—1. Identification and characterization. *Neuroscience* 10, 301–315.
- Grenhoff, J., Nisell, M., Ferré, S., Aston-Jones, G., Svensson, T.H., 1993. Noradrenergic modulation of dopamine cell firing elicited by stimulation of the locus coeruleus in the rat. *J. Neural Transm.-Gen.* 93, 11–25.
- Grenhoff, J., Svensson, T.H., 1993. Prazosin modulates the firing pattern of dopamine

- neurons in rat ventral tegmental area. *Eur. J. Pharmacol.* 233, 79–84.
- Gu, X., Hof, P.R., Friston, K.J., Fan, J., 2013. Anterior insular cortex and emotional awareness. *J. Comp. Neurol.* 521, 3371–3388.
- Guyenet, P.G., Stornetta, R.L., Bochorishvili, G., DePuy, S.D., Burke, P.G.R., Abbott, S.B.G., 2013. C1 neurons: the body's EMTs. *Am. J. Physiol. Regul. Integr. Comp. Physiol.* 305, R187–R204.
- Haber, S.N., Fudge, J.L., McFarland, N.R., 2000. Striatonigrostriatal pathways in primates form an ascending spiral from the shell to the dorsolateral striatum. *J. Neurosci.* 20, 2369–2382.
- Hajo's-Korcsok, E., Sharp, T., 1996. 8-OH-DPAT-induced release of hippocampal noradrenaline in vivo: evidence for a role of both 5-HT_{1A} and D1 receptors. *Eur. J. Pharmacol.* 314, 285–291.
- Hand, G.A., Hewitt, C.B., Fulk, L.J., Stock, H.S., Carson, J.A., Davis, J.M., Wilson, M.A., 2002. Differential release of corticotropin-releasing hormone (CRH) in the amygdala during different types of stressors. *Brain Res.* 949, 122–130.
- Hasegawa, H., Piacentini, M.F., Sarre, S., Michotte, Y., Ishiwata, T., Romain Meeusen, R., 2008. Influence of brain catecholamines on the development of fatigue in exercising rats in the heat. *J. Physiol.* 586, 141–149.
- Hawkins, R.A., O'Kane, R.L., Simpson, I.A., Viña, J.R., 2006. Structure of the blood-brain barrier and its role in the transport of amino acids. *J. Nutr.* 136, 218S–226S.
- Herreros, I., Verschure, P.F.M.J., 2013. Nucleo-olivary inhibition balances the interaction between the reactive and adaptive layers in motor control. *Neural Netw.* 47, 64–71.
- Heyes, M.P., Garnett, E.S., Coates, G., 1988. Nigrostriatal dopaminergic activity is increased during exhaustive exercise stress in rats. *Life Sci.* 42, 1537–1542.
- Hill, E.E., Zack, E., Battaglini, C., Viru, M., Viru, A., Hackney, A.C., 2008. Exercise and circulating cortisol levels: the intensity threshold effect. *J. Endocrinol. Invest.* 31, 587–591.
- Hilty, L., Langer, N., Pascual-Marqui, R., Boutellier, U., Lutz, K., 2011a. Fatigue-induced increase in intracortical communication between mid /anterior insular and motor cortex during cycling exercise. *Eur. J. Neurosci.* 34, 2035–2042.
- Hilty, L., Jäncke, L., Luechinger, R., Boutellier, U., Lutz, K., 2011b. Limitation of physical performance in a muscle fatiguing handgrip exercise is mediated by thalamo-insular activity. *Hum. Brain Mapp.* 32, 2151–2160.
- Höglund, E., Balm, P.H., Winberg, S., 2000. Skin darkening, a potential social signal in subordinate arctic charr (*Salvelinus alpinus*): the regulatory role of brain monoamines and pro-opiomelanocortin-derived peptides. *J. Exp. Biol.* 203, 1711–1720.
- Holloway, B.B., Stornetta, R.L., Bochorishvili, G., Erisir, A., Viar, K.E., Guyenet, P.G., 2013. Monosynaptic glutamatergic activation of Locus Coeruleus and other lower brainstem noradrenergic neurons by the C1 cells in mice. *J. Neurosci.* 33, 18792–18805.
- Holroyd, C.B., Coles, M.G.H., 2002. The neural basis of human error processing: reinforcement learning, dopamine, and the error-related negativity. *Psychol. Rev.* 109, 679–709.
- Hu, Y., Liu, X., Qiao, D., 2015. Increased extracellular dopamine and 5-hydroxytryptamine levels contribute to enhanced subthalamic nucleus neural activity during exhausting exercise. *Biol. Sport.* 32, 187–192.
- Jacks, D.E., Sowash, J., Anning, J., McGloughlin, T., Andres, F., 2002. Effect of exercise at three exercise intensities on salivary cortisol. *J. Strength Cond. Res.* 16, 286–289.
- Jacobs, I., Bell, D.G., 2004. Effects of acute modafinil ingestion on exercise time to exhaustion. *Med. Sci. Sports Exerc.* 36, 1078–1082.
- Jiang, Q.Y., Kawashima, H., Iwasaki, Y., Uchida, K., Sugimoto, K., Itoi, K., 2004. Differential effects of forced swim-stress on the corticotropin-releasing hormone and vasopressin gene transcription in the paraventricular division of the paraventricular nucleus of rat hypothalamus. *Neurosci. Lett.* 358, 201–204.
- Johansen, J.P., Fields, H.L., 2004. Glutamatergic activation of anterior cingulate cortex produces an aversive teaching signal. *Nat. Neurosci.* 7, 398–403.
- Kalinski, M.I., Dluzen, D.E., Stadulis, R., 2001. Methamphetamine produces subsequent reductions in running time to exhaustion in mice. *Brain Res.* 921, 160–164.
- Katsurabayashi, S., Kubota, H., Tokutomi, N., Akaike, N., 2003. A distinct distribution of functional presynaptic 5-HT receptor subtypes on GABAergic nerve terminals projecting to single hippocampal CA1 pyramidal neurons. *Neuropharmacology* 44, 1022–1030.
- Kawashima, H., Saito, T., Yoshizato, H., Fujikawa, T., Sato, Y., McEwen, B.S., Soya, H., 2004. Endurance treadmill training in rats alters CRH activity in the hypothalamic paraventricular nucleus at rest and during acute running according to its period. *Life Sci.* 76, 763–774.
- Kay, D., Marino, F.E., Cannon, J., Gibson, A.S.C., Lambert, M.I., Noakes, T.D., 2001. Evidence for neuromuscular fatigue during high-intensity cycling in warm, humid conditions. *Eur. J. Appl. Physiol.* 84, 115–121.
- Kent-Braun, J.A., Fitts, R.H., Christie, A., 2012. Skeletal muscle fatigue. *Compr. Physiol.* 2, 997–1044.
- Kitaoaka, R., Fujikawa, G.T., Miyaki, T., Matsumura, S., Fushiki, T., Inoue, K., 2010. Increased noradrenergic activity in the ventromedial hypothalamus during treadmill running in rats. *J. Nutr. Sci. Vitaminol.* 56, 185–190.
- Klass, M., Roelands, B., Lévénez, M., Fontenelle, V., Pattyn, N., Meeusen, R., Duchateau, J., 2012. Effects of noradrenaline and dopamine on supraspinal fatigue in well-trained men. *Med. Sci. Sports Exerc.* 44, 2299–2308.
- Klass, M., Duchateau, J., Rabec, S., Meussen, R., Roelands, B., 2016. Noradrenaline reuptake inhibition impairs cortical output and limits endurance time. *Med. Sci. Sports Exerc.* 48, 1014–1023.
- Knoch, D., Fehr, E., 2007. Resisting the power of temptations: the right prefrontal cortex and self-control. *Ann. N. Y. Acad. Sci.* 1104, 123–134.
- Kougias, P., Weakley, S.M., Yao, Q., Lin, P.H., Chen, C., 2010. Arterial baroreceptors in the management of systemic hypertension. *Med. Sci. Monit.* 16, RA1–RA8.
- Kouneher, F., Charron, S., Koehlin, E., 2009. Motivation and cognitive control in the human prefrontal cortex. *Nat. Neurosci.* 12, 939–947.
- Kringelbach, M.L., 2005. The orbitofrontal cortex linking reward to hedonic experience. *Nat. Rev. Neurosci.* 6, 691–702.
- Lambert, E.V., St. Clair Gibson, A., Noakes, T.D., 2005. Complex systems model of fatigue: integrative homeostatic control of peripheral physiological systems during exercise in humans. *Br. J. Sports Med.* 39, 52–62.
- Langfort, J., Barańczuk, E., Pawlak, D., Chalimoniuk, M., Lukacova, N., Marsala, J., Górski, J., 2006. The effect of endurance training on regional serotonin metabolism in the brain during early stage of detraining period in the female rat. *Cell. Mol. Neurobiol.* 26, 1327–1342.
- Lefebvre, H., Compagnon, P., Contesse, V., Delarue, C., Thuillez, C., Vaudry, H., Kuhn, J.M., 2001. Production and metabolism of serotonin (5-HT) by the human adrenal cortex: paracrine stimulation of aldosterone secretion by 5-HT. *J. Clin. Endocr. Metab.* 86, 5001–5007.
- Marcora, S., 2009. Perception of effort during exercise is independent of afferent feedback from skeletal muscles, heart, and lungs. *J. Appl. Physiol.* 106, 2060–2062.
- Marcora, S.M., 2010. Counterpoint: afferent feedback from fatigued locomotor muscles is not an important determinant of endurance exercise performance. *J. Appl. Physiol.* 108, 454–456.
- Marcora, S.M., Staiano, W., Manning, V., 2009. Mental fatigue impairs physical performance in humans. *J. Appl. Physiol.* 106, 857–864.
- Mason, J.W., Hartley, L.H., Kotchen, T.A., Mougey, E.H., Ricketts, P.T., Jones, L.G., 1973. Plasma cortisol and norepinephrine responses in anticipation of muscular exercise. *Psychosom. Med.* 35, 406–414.
- McCormick, D.A., 1992. Neurotransmitter actions in the thalamus and cerebral cortex. *J. Clin. Neurophysiol.* 9, 212–223.
- McMorris, T., 2016. Developing the catecholamines hypothesis for the acute exercise-cognition interaction in humans: lessons from animal studies. *Physiol. Behav.* 165, 291–299.
- McMorris, T., Davranche, K., Jones, G., Hall, B., Minter, C., 2009. Acute incremental exercise, performance of a central executive task, and sympathoadrenal system and hypothalamic-pituitary-adrenal axis activity. *Int. J. Psychophysiol.* 73, 334–340.
- McMorris, T., Hale, B.J., 2012. Differential effects of differing intensities of acute exercise on speed and accuracy of cognition: a meta-analytical investigation. *Brain Cogn.* 80, 338–351.
- Medford, N., Critchley, H.D., 2010. Conjoint activity of anterior insular and anterior cingulate cortex: awareness and response. *Brain Struct. Funct.* 214, 535–549.
- Meeusen, R., De Meirleir, K., 1995. Exercise and brain neurotransmission. *Sports Med.* 20, 160–188.
- Meeusen, R., Piacentini, M.F., De Meirleir, K., 2001a. Brain microdialysis in exercise research. *Sports Med.* 31, 965–983.
- Meeusen, R., Piacentini, M.F., Van Den Eynde, S., Magnus, L., De Meirleir, K., 2001b. Exercise performance is not influenced by a 5-HT reuptake inhibitor. *Int. J. Sports Med.* 22, 329–336.
- Meeusen, R., Roelands, B., 2017. Fatigue: Is it all neurochemistry? *Eur. J. Sports Sci.* <http://dx.doi.org/10.1080/17461391.2017.1296890>.
- Meeusen, R., Roeykens, J., Magnus, L., Keizer, H., De Meirleir, K., 1997. Endurance performance in humans: the effect of a dopamine precursor or a specific serotonin (5-HT_{2A/2C}) antagonist. *Int. J. Sports Med.* 18, 571–577.
- Meeusen, R., Thorré, K., Chaouloff, F., Sarre, S., De Meirleir, K., Ebinger, G., Michotte, Y., 1996. Effects of tryptophan and/or acute running on extracellular 5-HT and 5-HIAA levels in the hippocampus of food-deprived rats. *Brain Res.* 740, 245–252.
- Meeusen, R., Watson, P., Hasegawa, H., Roelands, B., Piacentini, M.F., 2006. Central fatigue the serotonin hypothesis and beyond. *Sports Med.* 36, 881–909.
- Mehta, J.P., Verber, M.D., Wieser, J.A., Schmit, B.D., Schindler-Ivens, S.M., 2009. A novel technique for examining human brain activity associated with pedaling using fMRI. *J. Neurosci. Meth.* 179, 230–239.
- Mejías-Aponte, C.A., Drouin, C., Aston-Jones, G., 2009. Adrenergic and noradrenergic innervation of the midbrain ventral tegmental area and retrorubral field: prominent inputs from medullary homeostatic centers. *J. Neurosci.* 29, 3613–3626.
- Mesulam, M.M., Mufson, E.J., 1982. Insula of the old world monkey. III: efferent cortical output and comments on function. *J. Comp. Neurol.* 212, 38–52.
- Miyashita, T., Williams, C.L., 2006. Epinephrine administration increases neural impulses propagated along the vagus nerve: role of peripheral beta-adrenergic receptors. *Neurobiol. Learn. Mem.* 85, 116–124.
- Moor, T., Mundorff, L., Bohringer, A., Philippsen, C., Langewitz, W., Reino, S.T., Schachinger, H., 2005. Evidence that baroreflex feedback influences long-term incidental visual memory in men. *Neurobiol. Learn. Mem.* 84, 168–174.
- Moore, C.I., 2004. Frequency-dependent processing in the vibrissa sensory system. *J. Neurophysiol.* 91, 2390–2399.
- Moore, K.E., Lariviere, E.W., 1964. Effects of stress and d-amphetamine on rat brain catecholamines. *Biochem. Pharmacol.* 13, 1098–1100.
- Morelli, M., Pinna, A., 2001. Interaction between dopamine and adenosine A2A receptors as a basis for the treatment of Parkinson's disease. *Neurosci. Sci.* 22, 71–72.
- Mravec, B., 2006. Possible involvement of the vagus nerve in monitoring plasma catecholamine levels. *Neurobiol. Learn. Mem.* 86, 353–355.
- Nee, D.E., D'Esposito, M., 2016. The hierarchical organization of the lateral prefrontal cortex. *eLife* 5, e12112. <http://dx.doi.org/10.7554/eLife.12112>.
- Newsholme, E.A., Acworth, I.N., Blomstrand, E., 1987. Amino acids, brain neurotransmitters and a functional link between muscle and brain that is important in sustained exercise. In: Benzi, G. (Ed.), *Advances in Biochemistry*. John Libbey, London, pp. 127–133.
- Noakes, T.D., 2012. Fatigue is a brain-derived emotion that regulates the exercise behavior to ensure the protection of whole body homeostasis. *Front. Physiol.* <http://dx.doi.org/10.3389/fphys.2012.00082>.
- Noakes, T.D., Marino, F., 2009. Point: maximal oxygen uptake is not limited by a central nervous system governor. *J. Appl. Physiol.* 102, 781–786.

- Noakes, T.D., St. Clair Gibson, A., Lambert, E.V., 2004. From catastrophe to complexity: a novel model of integrative central neural regulation of effort and fatigue during exercise in humans. *Br. J. Sports Med.* 38, 511–514.
- Ohiwa, N., Saito, T., Chang, H., Omoria, T., Fujikawa, T., Asada, T., Soya, H., 2006. Activation of A1 and A2 noradrenergic neurons in response to running in the rat. *Neurosci. Lett.* 395, 46–50.
- Olijslagers, J.E., Werkman, T.R., McCreary, A.C., Kruse, C.G., Wadman, W.J., 2006. Modulation of midbrain dopamine neurotransmission by serotonin, a versatile interaction between neurotransmitters and significance for antipsychotic drug action. *Curr. Neuropharmacol.* 4, 59–68.
- Onus, K., Cannon, J., Libert, L., Marino, F.E., 2016. Acute effects of a dopamine/norepinephrine reuptake inhibitor on neuromuscular performance following self-paced exercise in cool and hot environments. *J. Therm. Biol.* 60, 60–69.
- Pageaux, B., Lepers, R., Dietz, K.C., Marcora, S.M., 2014. Response inhibition impairs subsequent self-paced endurance performance. *Eur. J. Appl. Physiol.* 114, 1095–1105.
- Pageaux, B., Marcora, S.M., Rozand, V., Lepers, R., 2015. Mental fatigue induced by prolonged self-regulation does not exacerbate central fatigue during subsequent whole-body endurance exercise. *Front. Hum. Neurosci.* <http://dx.doi.org/10.3389/fnhum.2015.00067>.
- Pardridge, W.M., 1998. Blood–brain barrier carrier-mediated transport and brain metabolism of amino acids. *Neurochem. Res.* 23, 635–644.
- Parise, G., Bosman, M.J., Boecker, D.R., Barry, M.J., Tarnopolsky, M.A., 2001. Selective serotonin reuptake inhibitors: their effect on high-intensity exercise performance. *Arch. Phys. Med. Rehabil.* 82, 867–871.
- Paterson, S., Marino, F.E., 2004. Effect of deception of distance on prolonged cycling performance. *Percept. Mot. Skills* 98, 1017–1026.
- Piacentini, M.F., Meeusen, R., Buyse, L., De Schutter, G., De Meirleir, K., 2004. Hormonal responses during prolonged exercise are influenced by a selective DA/NA reuptake inhibitor. *Br. J. Sports Med.* 38, 129–133.
- Piacentini, M.F., Meeusen, R., Buyse, L., De Schutter, G., Kempnaers, F., Van Nijvel, J., De Meirleir, K., 2002. No effect of a noradrenergic reuptake inhibitor on performance in trained cyclists. *Med. Sci. Sports Exerc.* 34, 1189–1193.
- Podolin, D.A., Munger, P.A., Mazzeo, R.S., 1991. Plasma-catecholamine and lactate response during graded-exercise with varied glycogen conditions. *J. Appl. Physiol.* 71, 1427–1433.
- Pruessner, J.C., Champagne, F., Meaney, M.J., Dagher, A., 2004. Dopamine release in response to a psychological stress in humans and its relationship to early life maternal care: a positron emission tomography study using [¹¹C]raclopride. *J. Neurosci.* 24, 2625–2631.
- Puig, M.V., Celada, P., Diaz-Mataix, L., Artigas, F., 2003. In vivo modulation of the activity of pyramidal neurons in the rat medial prefrontal cortex by 5-HT_{2A} receptors: relationship to thalamocortical afferents. *Cereb. Cortex* 13, 870–882.
- Rauch, H.G.L., St. Clair Gibson, A., Lambert, E.V., Noakes, T.D., 2005. A signalling role for muscle glycogen in the regulation of pace during prolonged exercise. *Br. J. Sports Med.* 39, 34–38.
- Rinaman, L., 2011. Hindbrain noradrenergic A2 neurons: diverse roles in autonomic, endocrine, cognitive, and behavioral functions. *Am. J. Physiol. Regul. Integr. Comp. Physiol.* 300, R222–R235.
- Robertson, C.V., Marino, F.E., 2016. A role for the prefrontal cortex in exercise tolerance and termination. *J. Appl. Physiol.* 120, 464–466.
- Roelands, B., Goekint, M., Heyman, E., Piacentini, M.F., Watson, P., Hasegawa, H., Buyse, L., Pauwels, F., De Schutter, G., Meeusen, R., 2008a. Acute norepinephrine reuptake inhibition decreases performance in normal and high ambient temperature. *J. Appl. Physiol.* 105, 206–212.
- Roelands, B., Hasegawa, H., Watson, P., Piacentini, M.F., Buyse, L., De Schutter, G., Meeusen, R., 2008b. The effects of acute dopamine reuptake inhibition on performance. *Med. Sci. Sports Exerc.* 40, 879–885.
- Salamone, J.D., Correa, M., Mingote, S., Weber, S.M., 2003. Nucleus accumbens dopamine and the regulation of effort in food-seeking behavior: implications for studies of natural motivation, psychiatry, and drug abuse. *J. Pharmacol. Exp. Ther.* 305, 1–8.
- Salamone, J.D., Correa, M., Nunes, E.J., Randall, P.A., Pardo, M., 2012. The behavioral pharmacology of effort-related choice behavior: dopamine, adenosine and beyond. *J. Exp. Anal. Behav.* 97, 125–146.
- Samorajski, T., Marks, B.H., 1962. Localization of tritiated norepinephrine in mouse brain. *J. Histochem. Cytochem.* 10, 393–399.
- Santiago, A., Machado, A., Cano, J., 1995. 5-HT₃ receptor agonist induced carrier-mediated release of dopamine in rat striatum in vivo. *Br. J. Pharmacol.* 116, 1545–1550.
- Schoenbaum, G., Roesch, M.R., Stalnaker, T.A., Takahashi, Y.K., 2009. A new perspective on the role of the orbitofrontal cortex in adaptive behaviour. *Nat. Rev. Neurosci.* 10, 885–892.
- Schultz, W., 1998. The phasic reward signal of primate dopamine neurons. *Adv. Pharmacol.* 42, 686–690.
- Schultz, W., 2007. Multiple dopamine functions at different time courses. *Annu. Rev. Neurosci.* 30, 259–288.
- Seamans, J.K., Yang, C.R., 2004. The principal features and mechanisms of dopamine modulation in the prefrontal cortex. *Prog. Neurobiol.* 74, 1–57.
- Shadmehr, R., Krakauer, J.W., 2008. A computational neuroanatomy for motor control. *Exp. Brain Res.* 185, 359–381.
- Shen, H.-Y., Chen, J.-F., 2009. Adenosine A_{2A} receptors in psychopharmacology: modulators of behavior, mood and cognition. *Curr. Neuropharmacol.* 7, 195–206.
- Sherrington, C.S., 1948. *The Integrative Action of the Nervous System*. Cambridge University Press, Cambridge, UK.
- Shojaei, E.A., Farajov, A., Jafari, A., 2011. Effect of moderate aerobic cycling on some systemic inflammatory markers in healthy active collegiate men. *Int. J. Gen. Med.* 4, 79–84.
- Singer, T., Critchley, H.D., Preusschoff, K., 2009. A common role of insula in feelings, empathy and uncertainty. *Trends Cogn. Sci.* 13, 334–340.
- Soya, H., Mukai, A., Deocaris, C.C., Ohiwa, N., Chang, H., Nishijima, T., Fujikawa, T., Togashi, K., Saito, T., 2007. Threshold-like pattern of neuronal activation in the hypothalamus during treadmill running: establishment of a minimum running stress (MRS) rat model. *Neurosci. Res.* 58, 341–348.
- Spielberg, J.M., Miller, G.A., Warren, S.L., Engels, A.S., Crocker, L.D., Banich, M.T., Sutton, B.P., Heller, W., 2012. A brain network instantiating approach and avoidance motivation. *Psychophysiology* 49, 1200–1214.
- St. Clair Gibson, A., Baden, D.A., Lambert, E.V., Harley, Y.X., Hampson, D., Russell, V.A., Noakes, T.D., 2003. The conscious perception of the sensation of fatigue. *Sports Med.* 33, 167–176.
- St. Clair Gibson, A., Lambert, E.V., Rauch, L.H.G., Tucker, R., Baden, D.A., Foster, C., Noakes, T.D., 2006. The role of information processing between the brain and peripheral physiological systems in pacing and perception of effort. *Sports Med.* 36, 705–722.
- St. Clair Gibson, A., Noakes, T.D., 2004. Evidence for complex system integration and dynamic neural regulation of skeletal muscle recruitment during exercise in humans. *Br. J. Sports Med.* 38, 797–806.
- Strachan, A.T., Leiper, J.B., Maughan, R.J., 2004. Paroxetine administration failed [corrected] to influence human exercise capacity, perceived effort or hormone responses during prolonged exercise in a warm environment. *Exp. Physiol.* 89, 657–664.
- Suzuki, M., Matsuda, T., Asano, S., Sonboonthum, P., Takuma, K., Baba, A., 1995. Increase of noradrenaline release in the hypothalamus of freely moving rat by post-synaptic 5-hydroxytryptamine_{1A} receptor activation. *Br. J. Pharmacol.* 115, 703–711.
- Swart, J., Lamberts, R.P., Lambert, M.I., St. Clair Gibson, A., Lambert, E.V., Skowno, J., Noakes, T.D., 2009. Exercising with reserve: evidence that the central nervous system regulates prolonged exercise performance. *Br. J. Sports Med.* 43, 782–788.
- Szabo, S.T., Blier, P., 2001. Functional and pharmacological characterization of the modulatory role of serotonin on the firing activity of locus coeruleus norepinephrine neurons. *Brain Res.* 922, 9–20.
- Szatkowska, I., Bogorodzki, P., Wolak, T., Marchewka, A., Szeszkowski, W., 2008. The effect of motivation on working memory: an fMRI and SEM study. *Neurobiol. Learn. Mem.* 90, 475–478.
- Taylor, J.L., Gandevia, S.C., 2008. A comparison of central aspects of fatigue in sub-maximal and maximal voluntary contractions. *J. Appl. Physiol.* 104, 542–550.
- Timofeeva, E., Huang, Q., Richard, D., 2003. Effects of treadmill running on brain activation and the corticotropin-releasing hormone system. *Neuroendocrinology* 77, 388–405.
- Ulmer, H.-V., 1996. Concept of an extracellular regulation of muscular metabolic rate during heavy exercise in humans by psychophysiological feedback. *Experientia* 52, 416–420.
- Usher, M., Cohen, J.D., Servan-Schreiber, D., Rajkowski, J., Aston-Jones, G., 1999. The role of locus coeruleus in the regulation of cognitive performance. *Science* 283, 549–554.
- Van Bockstaele, E.J., Colago, E.E., Valentino, R.J., 1996. Corticotropin-releasing factor-containing axon terminals synapse onto catecholamine dendrites and may pre-synaptically modulate other afferents in the rostral pole of the nucleus locus coeruleus in the rat brain. *J. Comp. Neurol.* 364, 523–534.
- Varazzani, C., San-Galli, A., Gilardeau, S., Bouret, S., 2015. Noradrenaline and dopamine neurons in the reward/effort trade-off: a direct electrophysiological comparison in behaving monkeys. *J. Neurosci.* 35, 7866–7877.
- Velásquez-Martínez, M.C., Vázquez-Torres, R., Jiménez-Riveira, C.A., 2012. Activation of alpha₁-adrenoreceptors enhances glutamate release onto ventral tegmental area dopamine cells. *Neuroscience* 216, 18–30.
- Vogt, B.A., 2005. Pain and emotion interactions in subregions of the cingulate gyrus. *Nat. Rev. Neurosci.* 6, 533–544.
- Vytacil, J., Kayser, A., Miyakawa, A., D'Esposito, M., 2014. An approach for identifying brainstem dopaminergic pathways using resting state functional MRI. *PLoS One* 9 (1), e87109. <http://dx.doi.org/10.1371/journal.pone.0087109>.
- Waelti, P., Dickinson, A., Schultz, W., 2001. Dopamine responses comply with basic assumptions of formal learning theory. *Nature* 412, 43–48.
- Wang, M., Ramos, B.P., Paspalas, C.D., Shu, Y., Simen, A., Duque, A., Vijayraghavan, S., Brennan, A., Dudley, A., Nou, E., Mazer, J.A., McCormick, D.A., Arnsten, A.F.T., 2007. α_{2A}-Adrenoreceptors strengthen working memory networks by inhibiting cAMP-HCN channel signaling in prefrontal cortex. *Cell* 129, 397–410.
- Watson, P., Hasegawa, H., Roelands, B., Piacentini, M.F., Looover, R., Meeusen, R., 2005. Acute dopamine/noradrenaline reuptake inhibition enhances human exercise performance in warm, but not temperate conditions. *J. Physiol.* 565, 873–883.
- Watson, A.W.S., 1974. The relationship between tidal volume and respiratory frequency during muscular exercise. *Br. J. Sports Med.* 8, 87–90.
- Williams, S.M., Goldman-Rakic, P.S., 1998. Widespread origin of the primate mesofrontal dopamine system. *Cereb. Cortex* 8, 321–345.
- Williamson, J.W., McColl, R., Mathews, D., Ginsburg, M., Mitchell, J.H., 1999. Activation of the insular cortex is affected by the intensity of exercise. *J. Appl. Physiol.* 87, 1213–1219.
- Wilson, W., Maughan, R., 1992. Evidence for a possible role of 5-hydroxytryptamine in the genesis of fatigue in man: administration of paroxetine, a 5-HT re-uptake inhibitor, reduces the capacity to perform prolonged exercise. *Exp. Physiol.* 77, 921–924.
- Yanagita, S., Amemiya, S., Suzuki, S., Kita, I., 2007. Effects of spontaneous and forced running on activation of hypothalamic corticotropin-releasing hormone neurons in rats. *Life Sci.* 80, 356–363.